

The Minimal Architectural Ladder for Relational Computation: From Temporal Saliency to Binding and Composition in a Dynamical Neuron Model

Li Zexu

School of Physics and Astronomy, University of Leeds

correspondence: [correspondence email pending submission]

Final Revised Manuscript (September 2026)

Abstract

Modern computational neuroscience and deep learning routinely conflate five distinct computational primitives—memory, saliency weighting, content routing, semantic binding, and composition—and infer the emergence of the higher ones from increases in dynamical complexity alone. This paper replaces that inference with a constructive question: given a fixed dynamical substrate, which minimal organizational degree of freedom must be added to ascend from one primitive to the next? We answer the question inside a single dynamical unit, the Temporal Attention Neuron (TAN), whose scalar-input, scalar-gate surprise-weighted kernel was previously frozen (TAN-I) as a system that is historically dependent, geometrically responsive, and yet provably incapable of query-conditioned routing. Proceeding one degree of freedom at a time and auditing every step with counterfactual suites and negative-control chains, we establish six principal results. (i) Excitatory–inhibitory opponent topology suffices for nonmonotonic selective tuning but not for routing (Competition $\nRightarrow$ Routing; six layouts, FlipRate $\equiv$ 0.000). (ii) Two-dimensional vector query–key interaction unlocks routing capacity with an exact geometric law, FlipRate_{CF}(θ) = θ/π (132/132 Monte Carlo conditions), and realizes counterfactual winner flips in the running neuron (fixed history, query 3 $\rightarrow$ 4: margin +0.2587 $\rightarrow$ -0.1559; ensemble 0.4425 vs. benchmark 0.4445). (iii) Key–value decoupling yields soft binding, and the winner-take-all limit ($\gamma \rightarrow \infty$) yields exact, lossless delivery while leaving routing statistics unchanged—the first orthogonal axis: *routing selectivity* $\perp$ *readout discreteness*. (iv) A single head cannot simultaneously deliver two distinct bound operands to downstream algebra (Binding $\nRightarrow$ Composition); while a single head can evaluate symmetric pooling functions, two parallel channels plus one minimal two-input algebraic node are required to achieve zero composition error conditioned on correct binding, with all unconditional error attributable to upstream routing—the second orthogonal axis: *binding fidelity* $\perp$ *combiner error*. (v) Auditing address identifiability in multi-candidate retrieval ($N = 3$, 60,000 trials, 420,000 model evaluations) proves that scalar dot-product kernels mathematically cannot address intermediate keys (0/6,705 hits, 0.00%), whereas a one-dimensional metric distance kernel achieves 100.00% routing identical to two-dimensional vector rotation under hard selection, proving that metric compatibility rather than multidimensional rotation is the true mathematical boundary of content addressing. (vi) A pre-registered confirmatory audit of continuous dynamical carriers and discrete spiking readouts (Phase P1-B, 96,000 episodes, 192,000 queries) certifies that subthreshold inputs undergo analytical state collisions erasing history, baseline dynamical carriers fail prerequisite delivery floors (2.75%–2.90% joint accuracy), rendering memory comparisons floor-limited with memory redundancy unestablished, and single-unit spiking readouts exhibit severe transmission breakdown ($E_{\text{comp}} \in [2.21, 2.57]$, 82% abstention), certifying model-specific carrier insufficiency and point-process transmission limits while deferring internal composition dynamics. The result is a unified, stepwise mechanistic ladder of relational computation and an explicit map of the walls that separate its rungs.

1 Introduction: The Conflation Problem and the Architectural Stance

1.1 The conflation problem

A striking feature of the modern literature on attention, memory, and relational computation is that its central vocabulary is used interchangeably across levels that are causally distinct. A network that is “attentive” is described as routing information; a unit whose output is a weighted mixture of stored items is described as binding them; a system that can retrieve two stored values is described as composing them. Underneath this vocabulary sits a chain of five primitives—*memory*, *saliency weighting*, *content routing*, *semantic binding*, and *composition*—each of which imposes its own, strictly stronger, requirement on the substrate that realizes it. When these are conflated, the classic inferential error follows: observing the *lower* primitive, together with rich nonlinear dynamics, is taken as evidence for the *higher* one. Four examples of the conflation recur in our own field of view: (a) *history dependence* $\Rightarrow$ *attention routing*—a non-Markovian hidden state does not imply that a query can reorder candidate preferences; (b) *attention weighting* $\Rightarrow$ *semantic binding*—a softmax mixture over stored items delivers a convex combination, not a bound symbol; (c) *sharpening* $\Rightarrow$ *selection*—any finite temperature keeps every candidate weight strictly positive, so sharpening approaches, but never achieves, discrete delivery; and (d) *retrieval* $\Rightarrow$ *composition*—delivering two values into a pair is not the same as applying a deterministic algebraic relation to them.

This is not merely a semantic complaint. End-to-end training of monolithic networks makes the conflation unfalsifiable in practice: every stage of a black-box system co-varies under optimization, so post hoc attribution of a capability to a mechanism is, in general, impossible [3–5]. The binding problem of classical neuroscience is the canonical instance: synchrony, assembly formation, and indexing have all been proposed as binding mechanisms [12, 13], yet a demonstration that a candidate mechanism *binds*, as opposed to merely re-weights, is rarely separated from the demonstration that it *routes*. What the field lacks is not more powerful models but a *boundary map*: for each rung of the primitive ladder, the minimal structural condition under which the primitive becomes observable in a dynamical substrate, and a falsifiable test that separates it from the rungs below.

1.2 Thesis and methodological stance

The thesis of this paper is architectural rather than dynamical:

Computational capability is constrained not only by nonlinear dynamics, but by the organizational degrees of freedom available to the dynamics.

More dynamics—richer nonlinearities, larger state spaces, longer memory windows—does not, by itself, produce new computational primitives. It provides a richer *stage*; the primitives are provided by the *topological organization* of the computation. To make the thesis testable we adopt two disciplines that run through the entire study.

Radical minimalism. The substrate is a single dynamical unit—the Temporal Attention Neuron (TAN) of the frozen TAN-I archive—a leaky, spiking neuron whose input current is a surprise-gated, temporally attended mixture of its recent inputs (§2.3). It is deliberately the smallest system in which the conflation (a)–(d) can be exhibited, because every confound that a larger network would introduce (depth, width, training, shared substrates) is absent by construction.

The single-degree-of-freedom ladder. Each step of the study changes exactly one organizational degree of freedom and leaves the frozen dynamics, window, gates, and readouts untouched (§2.4). After each step, the new configuration is audited by counterfactual experiments—identical histories with only the query changed—and by negative-control chains in which the

preceding rung of the ladder is shown to fail the same test (§2.5). A rung is claimed only if the audit passes under the joint criteria, never from a single scalar error.

We state the resulting scope preemptively. This paper does not propose a network that beats any benchmark; it proposes a *constraint map* for dynamical computation. Its results are *sufficiency* statements within one explicitly frozen mechanism family, not necessity statements about brains or about all possible machines; and negative results are reported with the same status as positive ones. Readers expecting accuracy tables will find boundary tables instead; that is the intended product.

1.3 Why a single neuron is the right test bed

The TAN-I programme established the bottom two rungs of the ladder inside this substrate, and—decisively—located the first wall above them. In brief (full numbers and protocols in §3–§5): the scalar TAN is historically dependent and its membrane potential is not a sufficient Markov state (Phase 1: $N = 160,000$ collision experiments; state insufficiency lemma), so *memory* is realized. Its surprise-gated attention reshapes the event-locked response geometry of the system (Phase 2: effective rank $d_D \approx 2.089$ under attention versus ≈ 1.484 without), so *saliency weighting* is realized and is geometrically effective. Yet two negative probes froze the boundary: a temporal distractor task showed no robustness premium attributable to attention, and a formal binding audit—the query-conditioned identifiability audit, archived as TERMINATED / AUDIT_FAILED—recorded a counterfactual argmax flip rate of exactly 0.000: the scalar kernel $E_{t,i} = \beta W_q W_k S_t x_i$ can rescale candidate logits but can never reorder them, because its “query” is the non-negative scalar surprise cone $\mathcal{Q}_+ = [0, \infty)$. Scalar saliency, in other words, lies *below* routing on the ladder, and the gap is provable rather than empirical (the order-conservation theorem, §5). The TAN therefore leaves us exactly where a mechanism study should start: at a wall, with a clean statement of which side each frozen result belongs to.

1.4 Contributions and roadmap

The present paper (TAN-II, sprints 4.1–4.3-A) ascends the ladder one degree of freedom at a time and reports the following results.

1. **Competition without routing (§4).** Adding excitatory–inhibitory opponent topology to the frozen neuron produces nonmonotonic selective tuning by analytic construction—the E-I nonmonotonicity theorem—and the static opponent control shows the peak exists with *no* attention at all ($x^* = 1.00$, $h_E = 0.200$), severing the attribution of local tuning to attention. Attention is thereby redefined as dynamic gain and operating-range modulation. Across six layouts the counterfactual flip rate remains exactly 0.000: Competition $\nrightarrow$ Routing.
2. **The geometric boundary of routing (§5).** Two-dimensional vector query–key interaction unlocks routing capacity with an exact law, $\text{FlipRate}_{\text{CF}}(\theta) = \theta/\pi$ (132/132 Monte Carlo conditions; winner measures 1/2; the winner-variation statistic $F_d = 1/2$ constant across $d \in \{2, 3, 4, 8\}$ as a negative control). In the running neuron, changing only the query from 3 to 4 on a fixed history flips the attention winner with margins $+0.2587 \rightarrow -0.1559$ (closed-form crossing $Q^* = 3.7071$; ensemble flip rate 0.4425 vs. quadrature benchmark 0.4445), and key normalization is shown to be a mechanistic enabler, not a cosmetic control.
3. **From routing to binding, and to exactness (§6–§7).** Key–value decoupling yields soft binding whose value-transfer is winner-aligned and provably pairing-sensitive ($T_{\text{swap}} = -T$ to 10^{-12}), with a softness distance $D \approx 1.7$ – 1.8 inherited from softmax convexity. The winner-take-all limit $\gamma \rightarrow \infty$ achieves exact, lossless delivery while the routing statistics

(FlipRate ≈ 0.4445) remain exactly flat across the entire sharpening ladder—the first orthogonal axis: *routing selectivity* $\perp$ *readout discreteness*.

4. **From binding to composition (§8).** A single head is a bottleneck: it delivers one scalar symbol per instant and cannot simultaneously deliver two distinct bound operands to general downstream algebra, so binding alone cannot compose general arithmetic (Binding $\nrightarrow$ Composition; the three-error joint audit catches the $V_A = 0$ accidental-zero trap). Two parallel channels plus one minimal two-input algebraic node achieve composition with *zero* error conditioned on correct binding, over 3×10^4 Monte Carlo histories ($E[E_{\text{comp}}|\text{both-routed}] \equiv 0$, $\max = 0$), while all unconditional error is attributable to upstream routing—the second orthogonal axis: *binding fidelity* $\perp$ *combiner error*.

These results assemble into the unified mechanistic ladder of §9 and the graded claim ledger of §10. The through-line of the paper is that each rung corresponds to one *named* organizational degree of freedom—opponent topology, vector interaction, key–value decoupling, discrete selection, parallel channels, and an algebraic node—and that each rung can be assigned a falsifiable boundary test. More dynamics did not have to be added at any step; organization did.

The remainder of the paper is organized as follows. §2 gives the formal framework: axiomatic definitions of the five primitives, the frozen baseline model, the catalog of extension operators, and the audit methodology. §3 records the state and geometry baseline inherited from TAN-I. §4–§8 present the five rungs and their audits in ladder order. §9 assembles the master table, discusses the biological reading of each degree of freedom (opponent microcircuits, oscillatory phase, dendritic subunits, axonal threshold integration), and states the general discussion. §10 closes with the claim ledger and the reproducibility statement.

2 Formal Framework: Five Primitives and the Frozen Baseline

This section fixes the definitions that the remainder of the paper uses without further negotiation: the five computational primitives (§2.2), the frozen dynamical baseline they are tested against (§2.3), the catalog of single-degree-of-freedom extensions that ascend the ladder (§2.4), and the audit methodology by which every rung is either established or rejected (§2.5). All quantitative constants in this section are taken verbatim from the frozen archives (TAN-I and TAN-II sprints 4.1–4.3-A); none are free.

2.1 Notation and conventions

Time is discrete, $t \in \mathbb{N}$. Inputs are scalar amplitudes $x_t \in \mathbb{R}$; the temporal window is the ordered W -tuple $X_t = (x_{t-W+1}, \dots, x_t)$ with zero padding at onset. We write $[z]_+ = \max(z, 0)$ for rectification, $\Theta(\cdot)$ for the Heaviside step, $\sigma(z) = 1/(1+e^{-z})$ for the logistic function, and $\text{softmax}_i(\mathbf{e})$ for the normalized exponential map. In Sprint 4.2-B/C/D the standard max-stabilized form $\alpha_i = \exp(e_i - \max_k e_k) / \sum_j \exp(e_j - \max_k e_k)$ is evaluated directly without offset ($\sum_i \alpha_i = 1.0$ exactly), whereas the baseline model implementation (`tan.py`) added a numerical stabilizer $\delta = 10^{-9}$ to the denominator. Alternative mappings that induce exact sparsity without infinite gain, such as `sparsemax` [7], are discussed as theoretical comparisons. Candidate sets are always finite. A *counterfactual pair* is a pair of experimental conditions that share the identical history content and differ only in the query; a *negative-control chain* is a sequence of models M_0, M_1, M_2 in which each model adds exactly one organizational degree of freedom and the preceding model is required to fail the criterion that the following model is required to pass.

2.2 The five primitives, operationally defined

The paper’s object is the chain Memory $\rightarrow$ Saliency $\rightarrow$ Routing $\rightarrow$ Binding $\rightarrow$ Composition. Each primitive is defined by an operational criterion that a frozen experiment can pass or fail;

the criteria are the ones actually used in the audits of §4–§8. The chain is constructive: in the mechanism family tested here, each later primitive presupposes the earlier ones, while the central negative results of this paper show that none of the reverse implications holds.

P1 — Memory.

Definition. A system has memory relative to a candidate state variable if its future evolution depends causally on aspects of the input history that are not recoverable from that candidate state. *Criterion* (TAN-I Phase 1, §3): the one-dimensional scalar membrane potential h_t is not state-sufficient; the projection $(h_t, X_t) \mapsto h_t$ is not lumpable. Two histories with equal membrane potentials but different window contexts produce different downstream trajectories (quantified by pairwise separation K and post-reset divergence). This demonstrates state non-sufficiency of scalar h_t ; it does not imply that the complete Markovian state (h_t, X_t) violates Markovianity, nor that Markovian dynamics imply state-space contraction.

P2 — Saliency weighting.

Definition. A system performs saliency weighting if its readout over a fixed candidate set preserves relative ranking across all active query levels. Under standard continuous softmax (whose support is everywhere strictly positive, in contrast to sparse activation functions such as sparsemax [7]), the defining signature of scalar modulation is not support-set invariance, but rather *ranking invariance*: for all $S > 0$, the induced ordering of attention weights $\alpha(Q)$ is strictly invariant, and $\operatorname{argmax}_i \alpha_i(Q)$ is independent of query content. *Criterion*: Rank-1 kernel factorizability $E(Q, K) = q(Q) \cdot k(K)$ is a sufficient condition ensuring that positive scalar query modulation acts purely as a monotonic gain on logit differences,

P3 — Content routing.

Definition. A system routes if there exist queries Q_A, Q_B and a fixed history such that $\operatorname{argmax}_i R(Q_A, K_i) \neq \operatorname{argmax}_i R(Q_B, K_i)$. In multi-candidate memory ($N \geq 3$), true content addressing requires selective identifiability: for every candidate key K_t , there exists a query making K_t the unique maximum. While separable dot-product kernels $E(Q, K) = QK$ in $\mathbb{R}$ can at most invert global order between the extreme elements without addressing intermediate keys (Proposition 5.4), metric compatibility kernels $E(Q, K) = -(Q - K)^2$ achieve universal addressability even in $d = 1$ (Proposition 5.3). *Criterion*: the counterfactual flip rate $\text{FlipRate}_{\text{CF}}$ is nonzero on a fixed-history ensemble; capacity (the geometric existence of flips) and realization (the running model actually flipping) are reported separately (§5).

P4 — Semantic binding.

Definition. A system binds if role/address (key) and content entity (value) are decoupled and the output transmits the value of the *selected* key, following the (key,value) association rather than key identity, position, or amplitude. *Criterion*: the value-swap counterfactual is antisymmetric, $T_{\text{swap}} = -T$ exactly, and the softness distance $D = |C - v_{\text{winner}}|$ is measured; hard binding is the limit $D = 0$.

P5 — Composition.

Definition. A system composes if it applies a deterministic algebraic relation f to at least two independently retrieved bound values, $y = f(V_A, V_B)$. *Criterion*: the three-error joint audit of §2.5 (binding errors and composition error simultaneously ≈ 0) together with the counterfactual suite of §8 (query swap, value swap, key swap, channel-output swap, the zero-value trap, and the binding-preserving, binding-breaking, and relational controls); conditioned on correct binding the composition error must be exactly zero (the binding-combiner separation test).

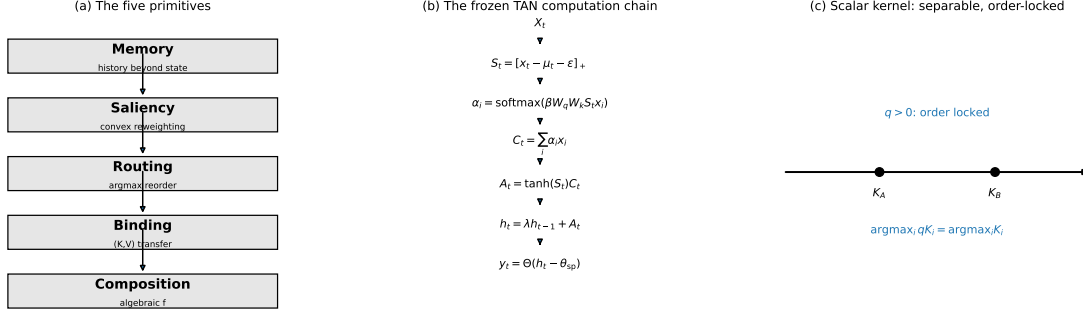


Figure 1: The conceptual taxonomy and the frozen TAN architecture. (a) The five primitives of §2.2 as a constructive chain. (b) The frozen TAN-I computation chain (§2.3). (c) The degenerate geometry of the scalar kernel: the non-negative query cone $\mathcal{Q}_+$ leaves the candidate ordering locked to the key magnitudes (Theorem 5.1).

2.3 The frozen baseline: the scalar Temporal Attention Neuron

The substrate is the TAN-I Temporal Attention Neuron, frozen before this study and never retrained or retuned here. Its computation chain is

$$\mu_t = \frac{1}{W} \sum_{k=0}^{W-1} x_{t-k}, \quad S_t = [x_t - \mu_t - \varepsilon]_+, \quad (1)$$

$$E_{t,i} = \beta W_q W_k S_t x_{t-i}, \quad \alpha_{t,i} = \frac{\exp(E_{t,i})}{\sum_j \exp(E_{t,j}) + \delta}, \quad (2)$$

$$C_t = \sum_i \alpha_{t,i} x_{t-i}, \quad A_t = \tanh(S_t) C_t, \quad (3)$$

$$h_t = \lambda h_{t-1} + A_t, \quad y_t = \Theta(h_t - \theta_{\text{sp}}), \quad h_t \leftarrow 0 \text{ on spike}. \quad (4)$$

Three properties of this chain carry the whole study and must be kept in view. (i) The query of the kernel (2) is the non-negative scalar surprise S_t (rectified by (1) and multiplied by positive constants), so the query domain is the closed cone $\mathcal{Q}_+ = [0, \infty)$; the kernel is therefore of the separable form (query scalar) $\times$ (key scalar), which pins the attention argmax to the key magnitudes regardless of the query—this is the exact content of the order-conservation theorem (§5) and the reason TAN-I froze at FlipRate = 0.000. (ii) The readout (3) is a convex combination, so the delivered “value” is always a mixture; discreteness is not available to this substrate until a selection operator is added (§7). (iii) The membrane update (4) contains the discrete spike-and-reset map; the system is piecewise discontinuous and must not be treated as a smooth autonomous dynamical system (§4’s conditional phase planes are labeled accordingly).

Table 1 is the frozen parameter inventory. Provenance columns refer to the frozen archive; the TAN-II ladder re-freezes the TAN-I values wherever it does not add a new degree of freedom, and every deviation is pre-registered (for example, the default spike threshold of the E-I stages is $\theta_{\text{sp}} = 3.0$, chosen pre-run to keep the tuning criterion decisive, with the TAN-I value 0.5 retained in the threshold sweep).

2.4 The catalog of extension operators

Each rung of the ladder is one of the following operators, applied to the baseline and left unchanged afterwards. No rung changes anything except the named operator.

Symbol	Frozen value	Meaning	Provenance
W	5	temporal win-dow	TAN-I
λ	0.5	membrane leak	TAN-I
θ_{sp}	0.5 (TAN-I); 3.0 default in TAN-II (sweep $\{0.5, 1.0, 3.0, 6.0\}$)	spike threshold	TAN-I / 4.1
$\varepsilon, \varepsilon_I$	0, 0.3	surprise dead zones	TAN-I / 4.1
β, W_q, W_k	1.0, 2.0, 1.0	kernel constants ($c = \beta W_q W_k = 2.0$)	TAN-I
δ	10^{-9}	softmax offset	TAN-I
(w_{EI}, w_{IE}, w_{II})	(0.8, 1.0, 0.2)	Dale-signed couplings	4.1
$(w_E, w_I, \theta_E, \theta_I)$	(0.5, 1.0, 0.4, 1.0)	static opponent thresholds (C1)	4.1
ω	0.4	query-direction rotation	4.2-B
(Q_A, Q_B)	(3.0, 4.0)	counterfactual queries (single-query)	TAN-I/4.2
(Q_1, Q_2)	(4.5, 5.3)	dual-query window	4.3-A
(V_A, V_B) scenarios	(5, 9), (7, -2), (2, 6)	canonical values	4.2-C/4.3-A
γ ladder	$\{1, 2, 5, 10, 100, \infty\}$	WTA sharpening	4.2-D
Seeds / N / grid	20260904-06; 10^4 ; 2001^2	Monte Carlo quadrature	4.2-4.3

Table 1: Frozen parameter inventory. All values are pre-registered; the provenance column names the frozen sprint. The spike threshold column shows the two frozen conventions and the sweep that audits their consequences.

O1 — Opponent competition (Sprint 4.1).

Two cells with membrane potentials $h_E(t) = \lambda_E h_E(t-1) + A_E(t) - w_{EI} A_I(t)$ and $h_I(t) = \lambda_I h_I(t-1) + w_{IE} A_E(t) - w_{II} A_I(t)$, where A_E, A_I are the per-branch drives (attention-gated contexts in the attended models; static thresholded currents $[w_E x_t - \theta_E]_+$, $[w_I x_t - \theta_I]_+$ in the static control C1) and all couplings obey Dale’s sign structure $w_{ab} \geq 0$. The four-model ablation ladder $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ of §4 instantiates this operator.

O2 — Vector query–key interaction (Sprint 4.2-B).

Keys are embedded as unit vectors $\hat{\varphi}_d(x) = p_d(x)/\|p_d(x)\|$ with $p_d(x) = (x, x^2, \dots, x^d)$ and $\hat{\varphi}_d(0) = 0$; the query is the rotating vector $Q_t = c S_t \hat{u}(\theta_t)$ with angle $\theta_t = \omega S_t$ and $\hat{u}(\theta) = (\cos \theta, \sin \theta, 0, \dots)$; the kernel becomes $E_{t,i} = Q_t^\top K_i$. Only the representation changes: window, surprise gate, dynamics, and readouts are frozen.

O3 — Key–value decoupling (Sprint 4.2-C).

Each slot carries a pair (k_i, v_i) : keys drive the attention, values are transmitted, $C_t = \sum_i \alpha_{t,i} v_i$. Query slots carry $v = 0$ and are never candidates.

O4 — Readout discreteness (Sprint 4.2-D).

The selection function $\alpha_i(\gamma) = \text{softmax}_i(\gamma \cdot \text{logit}_i)$ with $\gamma \in [1, \infty]$; $\gamma = \infty$ is the one-hot winner-take-all.

O5 — Parallel channels and the combiner (Sprint 4.3-A).

The frozen window is replaced by the dual-query window $[A, 0, B, Q_1, Q_2]$ with $(Q_1, Q_2) =$

(4.5, 5.3) (the direction-targeting derivation is frozen in the pre-registration); channel $c \in \{1, 2\}$ is keyed to slot $3 + c$ with the shared window mean, and a two-input node computes $y = f(\hat{C}_1, \hat{C}_2)$ with $f \in \{+, -\}$.

2.5 Audit methodology and discipline

Every rung is established or rejected by the following machinery, which is itself part of the frozen protocol and is reported in §10 together with the claim ledger.

Counterfactual suites. A result is claimed only if it survives the counterfactual manipulations of its rung: query swap, value swap, key swap, channel-output swap, position randomization, amplitude randomization, the single-query controls, and the leakage checks ($Q \cap V = \emptyset$; queries are never candidates; equal-norm keys). Subtraction provides the non-commutative operator whose swap responses are antisymmetric, addition the commutative one whose swap responses are invariant—both predictions are tested, so that correct commutative behavior can never be scored as a false negative.

The three-error joint audit. Composition success requires $E_{\text{bind},A} \approx 0$ and $E_{\text{bind},B} \approx 0$ and $E_{\text{comp}} = |y - f(V_A, V_B)| \approx 0$ simultaneously. The joint requirement exists to exclude the accidental-zero class of false positives exemplified by the $V_A = 0$ trap of §8, in which the composition error vanishes while a binding error remains.

Negative-control chains and orthogonal axes. Each positive claim is paired with the demonstration that the previous rung fails the same criterion (M_0 fails composition, M_1 binds but does not compose, M_2 composes). Two orthogonal axes are formalized from these chains: *routing selectivity* $\perp$ *readout discreteness* (§7: the WTA ladder changes discreteness while the flip rate stays exactly flat) and *binding fidelity* $\perp$ *combiner error* (§8: the combiner adds zero error conditioned on correct binding).

Statuses and stop rules. Every check receives an explicit PASS/FAIL/INCONCLUSIVE. Discrepancies with frozen predictions trigger the stop-and-debug procedure: stop, independently recompute, record the cause in a dated amendment that preserves the original prediction verbatim, and only then continue. The six amendments of this programme (AM1–AM6: the E4(iii) closed-form value, the unnormalized-control expectation, the corrected fixed-point constants, the E-I drive-transfer value, the readout-softness monotonicity, and the canonical routing margins) are all recorded this way; none altered a model, query, seed, or threshold.

Determinism and statistical criteria. All experiments are deterministic under the fixed seeds 20260904–20260906 ($N = 10^4$ per seed); analytic benchmarks are deterministic 2001² quadratures computed from the frozen model. Probability estimates are accepted when $|\hat{p} - p_{\text{bench}}| \leq 3 \text{ SE} + 0.002$ and mean-value estimates when $|\hat{m} - m_{\text{bench}}| \leq 3 \text{ SE}_{\text{mean}} + 0.01$; structural identities (swap antisymmetry, position invariance, conditioned composition error) are asserted at machine precision (10^{-12}).

3 From State to Geometry: Representation Without Selectivity

This section records the two frozen TAN-I results that anchor the bottom of the ladder: the scalar TAN has genuine *memory* (its state is not Markov-sufficient), and its attention genuinely *reorganizes* the event-locked response geometry—yet neither property grants the unit any content selectivity. The numbers are quoted verbatim from the frozen TAN-I archive; the section proves nothing new and re-runs nothing.

3.1 The state question: memory as non-Markovianity

The natural-collision experiment of TAN-I Phase 1 asks whether the pair (h_t, X_t) can be reduced to h_t : whether two histories that produce the same instantaneous membrane potential are thereafter indistinguishable. Over $N = 160,000$ simulated histories, four model variants—B1

(leaky integrate-and-fire), B2 (LIF with a fixed delay line), B3 (TAN without attention, $\alpha_i \equiv 1/W$), B4 (the full scalar TAN)—and the input control C0 were scanned for *natural collisions*: pairs of histories whose membrane states coincide while their window contexts differ. For each pair a state-separation statistic K measures how far the two futures diverge under the identical future input.

The frozen summary is: mean K is 0.5 *exactly* for B1 (maximum deviation 1.8×10^{-8} , the Markov bound), 4.47×10^3 [95% CI $3.26\text{--}5.92 \times 10^3$] for B2, 1.235×10^4 [$7.78 \times 10^3\text{--}1.83 \times 10^4$] for B3, and 2.047×10^4 [$1.12\text{--}3.57 \times 10^4$] for B4 (with corresponding medians of 0.5, 895.5, 1369.1, and 2947.7), with 1,500 pairs analysed per model; the fraction of pairs exceeding the Markov scale $K > 10^3$ is 0% / 46.7% / 56.7% / 73.7%, and the post-reset Markov-violation rate is 0% / 0.6% / 33.1% / 32.2%. Two controls certify the attribution: with the surprise gate closed, the B4 median collapses to the Markov value 0.5 (the memory lives in the gated context, not in the leak), and a re-synchronisation test on $n = 843$ fully resynchronised pairs shows the LIF remaining synchronised while the TAN variants re-bifurcate. The state-separation statistic correlates with the input-driven activity difference (Pearson $r(\text{JSD}, \Delta A) = 0.87\text{--}0.89$, $p < 10^{-300}$).

We state the frozen consequence as a lemma.

Lemma 3.1 (State insufficiency of the scalar TAN). *The projection $\pi : (h_t, X_t) \mapsto h_t$ is not lumpable for the frozen scalar TAN (B4): there exist histories $X_t \neq X'_t$ with $h_t = h'_t$ whose conditional futures differ. Equivalently, the effective state of the unit includes history content beyond the membrane potential; the system must be treated as history-dependent throughout, and every “phase portrait” constructed below is conditional on a frozen history context.*

Lemma 3.1 certifies the first rung: the substrate has memory (primitive P1 of §2.2). It also licenses the conditional-vector-field methodology used in §4: nothing downstream may pretend the scalar unit is an autonomous low-dimensional system.

3.2 The geometry question: event-locked effective dimension

TAN-I Phase 2 asked the next question in ladder order: granted memory and saliency weighting, does attention *reorganize* the unit’s response space? The frozen protocol locks the response to an event and measures an effective response dimension d_D of the event-locked fluctuation covariance, together with its log-trace $\log \text{Tr}$ as a scale statistic (seeds 20260904–06, $T = 5,000$, $n = 374$ events). The frozen event-velocity values are: B4 (full TAN) at its content-rich frame zF reaches $d_D = 2.089$ [95% CI $1.87\text{--}2.30$], versus B3 (no attention) $d_D = 1.484$ [$1.42\text{--}1.53$]; the plain LIF B1 and the input control C0 sit at 1.674 and 1.055. The amplitude-control ladder (raw / stratified / residual) gives B4-zF 2.089 / 2.250 / 2.280, and the B4–B3 contrast survives all three amplitude controls (+0.605 / +0.742 / +0.782). In scale units the event log-trace is 3.58 nats for B4-zF versus 2.48 nats for B3-zF (absolute difference +1.09 nats); the frozen audit’s paired delta statistic is $\Delta \log \text{Tr}(\text{B4} - \text{B3}, \text{zF}) = 1.355$ nats. The effect peaks at lag $\tau \approx 4\text{--}6$ (content-rich windows, peak $d_D \approx 2.89$), not at the first pulse, and decays with an e-folding time $\tau_e = 5.4$ [4.6–6.3] for B4 versus 8.5 [6.1–10.6] for B3: attention sharpens and *shortens* the event-locked response.

One negative control is decisive for the ladder. In the shared frame where B1 and B4 are compared on equal footing, B4 is *not* above B1—the attention gain is a reorganization relative to the attentionless TAN, not a dimension premium over the bare LIF. We carry this asymmetry forward: it is the first instance of the paper’s rule that a geometric change must not be read as a capability change without a boundary test.

3.3 Interpretation discipline: effective rank is not intrinsic dimension

The frozen archive is explicit about what Phase 2 does *not* license. d_D , the log-trace, and the spectra are estimator-level descriptions of local fluctuation covariance; they are *not* intrinsic manifold dimensions, and no capacity conclusion follows from them. Attention reshapes the event-locked response geometry—it does not, by that fact, select content. The correct summary of the bottom two rungs is therefore: memory (Lemma 3.1) and saliency weighting are realized, their joint effect on the response space is real and measurable, and *selectivity is absent*: nothing in §3 reorders any candidate under any query. The wall that separates this state of affairs from routing is the subject of §4–§5.

4 Competition Without Routing: The E-I Necessity Counterexample

The first candidate degree of freedom above scalar saliency is opponent competition: an excitatory–inhibitory microcircuit with Dale’s sign discipline. This section reports the frozen Sprint 4.1 ablation ladder $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ and its two outcomes: opponent topology suffices, by an analytic open-set theorem, for nonmonotonic selective tuning—*without any attention at all*; and it does nothing whatsoever to the routing boundary.

4.1 The opponent mechanism and its open-set theorem

Consider the static opponent readout on a scalar input $x \in [0, x_{\max}]$,

$$R(x) = [w_E x - \theta_E]_+ - g[w_I x - \theta_I]_+, \quad w_E, w_I, \theta_E, \theta_I, g \geq 0, \quad (5)$$

the instantaneous (pre-leak) form of the two-cell circuit used in §4.3. Its three regimes have slopes 0 (silence, $x < \theta_E/w_E$), $w_E > 0$ (excitation only), and $w_E - g w_I$ (both active). The frozen inequalities single out the regime in which the second slope is negative.

Theorem 4.1 (E-I nonmonotonicity). *Let $U = \{(w_E, w_I, \theta_E, \theta_I, g) : \theta_E/w_E < \theta_I/w_I < x_{\max}, g w_I > w_E\}$. For every parameter point in the open set U , the readout (5) has a strict local maximum at $x^* = \theta_I/w_I$: the left slope is $w_E > 0$ and the right slope is $w_E - g w_I < 0$. The maximum is strict and persists under the leaky embedding $h^*(x) = R(x)/(1 - \lambda)$ with $0 \leq \lambda < 1$.*

Proof. On $(\theta_E/w_E, \theta_I/w_I)$ only the first branch is active, so $R(x) = w_E x - \theta_E$ with slope $w_E > 0$; on $(\theta_I/w_I, x_{\max})$ both branches are active and $R(x) = (w_E - g w_I)x - \theta_E + g \theta_I$ with slope $w_E - g w_I < 0$ by the second inequality of U . Continuity at x^* follows from $[w_I x^* - \theta_I]_+ = 0$, so R is a continuous, piecewise-linear map that rises into x^* and falls away from it: a strict local maximum. Division by the positive constant $1 - \lambda$ preserves the sign of every slope. $\square$

With the frozen constants $(w_E, w_I, \theta_E, \theta_I, g) = (0.5, 1.0, 0.4, 1.0, 0.8)$ the conditions read $0.8 < 1.0 < 3.0$ and $0.8 \cdot 1.0 > 0.5$, the peak sits at $x^* = 1.00$ with height $h_E^* = 0.200$, and the leaky slopes are $+1.0$ and -0.6 (rising slope $w_E/(1 - \lambda)$, falling slope $(w_E - g w_I)/(1 - \lambda)$, $\lambda = 0.5$).

Remark. The theorem is the paper’s first necessity counterexample: it certifies that a local tuning peak exists in a configuration with *zero attention*, zero surprise gating, and zero temporal organization. Any later observation of nonmonotone selectivity inside an attended configuration therefore *cannot* be attributed to attention unless the attended rungs change the peak itself.

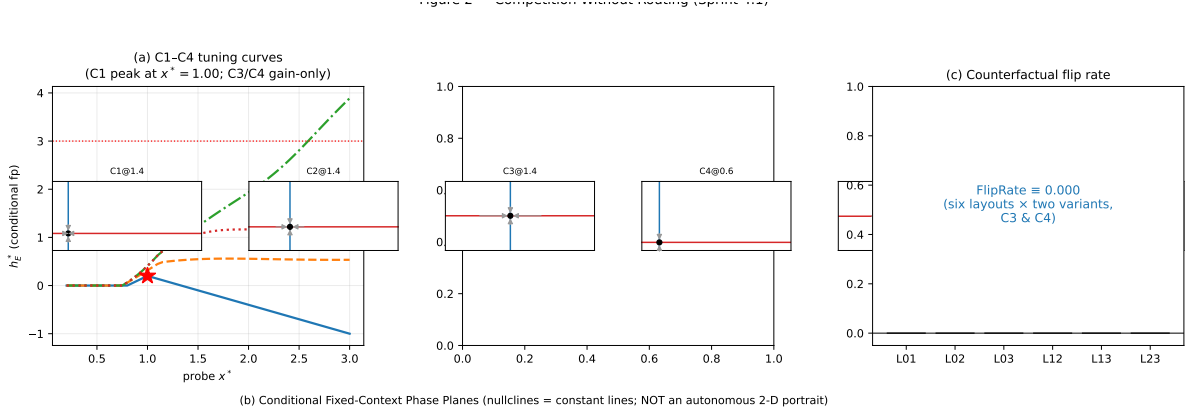


Figure 2: Competition without routing (Sprint 4.1). (a) The C1–C4 conditional-fixed-point tuning curves: only the static opponent model C1 exhibits a strict local maximum ($x^* = 1.00$, $h_E = 0.200$); the attended rungs C2–C4 only reshape gain. (b) Conditional fixed-context phase planes: the conditional nullclines are the constant lines $h_E = h_E^*$, $h_I = h_I^*$; they are *not* an autonomous two-dimensional portrait. (c) The counterfactual flip rate is exactly 0.000 across all six layouts and both variants.

4.2 The C1–C4 ablation ladder: what attention actually adds

The frozen four-rung ladder instantiates the operator O1 of §2.4 inside the conditional fixed-point readout of Lemma 3.1: C1, the static E-I LIF (drives $[w_E x_t - \theta_E]_+$, $[w_I x_t - \theta_I]_+$, no surprise, no window attention); C2, the TAN with surprise gating and forced uniform attention ($\alpha_i \equiv 1/W$); C3, asymmetric attention (E-branch attends, I-branch integrates uniformly); C4, the full classical E-I TAN (both branches with independent surprise and attention). The pre-registered monotonicity criterion P1 (finite-difference sign flip plus curvature, tolerances $10^{-3}/2 \times 10^{-3}/5 \times 10^{-3}$, spike adjacency excluded) was applied to the conditional fixed point $h_E^*(x^*)$ over the probe range $x^* \in [0.2, 3.0]$ (57 points, $\delta = 0.05$), and every prediction was frozen before the run.

The frozen verdicts: C1 **PASS** (peak at $x^* = 1.00$, height 0.200, prominence 0.100, curvature -0.11 , post-peak suppression slope -0.600 with $R^2 = 1.0$, maximal suppression 1.200); C2 **FAIL** (climb-then-plateau: maximum 0.5595 at $x^* = 1.80$ decaying only to 0.5344 at 3.0, a 0.025 plateau modulation below the curvature tolerance); C3 **FAIL** (monotone climb to 3.8897 at 3.0, spike onset 2.60, 15.8% of the scan in the spiking regime); C4 **FAIL** (monotone climb to 1.3592 at 3.0, subthreshold throughout). The episodic-protocol cross-checks reproduce every verdict, and the six-layout robustness sweep reproduces them layout by layout (C1 PASS, C2–C4 FAIL everywhere). The prediction table was hit 4/4.

The causal reading is forced by the ladder and is one of the paper’s central re-definitions. *Nonmonotone selectivity comes from the opponent threshold structure (C1), not from attention*: the attended rungs C2–C4 do not produce a single strict peak under the frozen criteria. What attention demonstrably does at each rung is gain and operating-range modulation: the surprise gate rectifies and saturates (C1’s negative inhibitory tail disappears, replaced by the C2 plateau); E-only attention amplifies gain until the unit enters the spiking regime (C3); and symmetric E/I attention re-reads the probe through the inhibitory branch and pulls the net response back into the subthreshold band (C4). Attention is thus a *dynamic gain and operating-point regulator* in this substrate—not a source of selectivity, and not a router.

4.3 Conditional phase planes, honestly labeled

For completeness the frozen sprint also froze the conditional vector fields: for a frozen history context X^* and a frozen probe x^* the drives $A_E(X^*, x^*)$, $A_I(X^*, x^*)$ are constants, so the

subthreshold flow is the affine contraction $h_a(t) = \lambda_a h_a(t-1) + B_a(X^*, x^*)$ with conditional fixed points $h_E^* = B_E/(1-\lambda_E)$, $h_I^* = B_I/(1-\lambda_I)$ and conditional nullclines that are the *constant lines* $h_E = h_E^*$, $h_I = h_I^*$. Every figure in the frozen archive labels these “Conditional Fixed-Context Phase Plane” and states explicitly that they are *not* an autonomous two-dimensional phase portrait; different probe values translate the affine field rather than deform an intrinsic topology, and any probe whose fixed point crosses the spike threshold is flagged as a spiking regime. No claim of two-dimensional autonomous dynamics is made anywhere in this paper.

4.4 The routing boundary survives competition

The counterfactual criterion P2 (fixed history, queries $Q_A = 3.0$, $Q_B = 4.0$, winner = attention argmax over history candidates, two variants) was run across all six event layouts. The frozen outcome is FlipRate $\equiv 0.000$ in every layout, every variant, and every attended model (C3, C4; C1/C2 have no candidate structure and were pre-registered as not applicable), while the response magnitude is query-sensitive ($\Delta R = +1.827$ for C3, $+0.281$ for C4). The spike threshold sweep ($\theta_{\text{sp}} \in \{0.5, 1.0, 3.0, 6.0\}$) documents the spiking onsets (e.g. C3 at 2.60 for the default 3.0; C2/C3/C4 at 1.25/1.05/1.05 for the TAN-I threshold 0.5) and shows that the monotonicity verdicts live entirely in the subthreshold region and are threshold-robust. The structural reason for the flat zero is recorded for later use: the winner is decided by the kernel’s argmax *before* any E-I dynamics, and opponent competition acts on the scalar aggregate $C = \sum_i \alpha_i v_i$, which cannot reorder candidates. The boundary to be crossed is therefore not dynamical but representational, which is precisely what the next rung changes.

$$\boxed{\text{E-I Competition} \not\Rightarrow \text{Content Routing}}$$

5 The Geometric Boundary of Routing: Hyperplane Bisection in $d = 2$

Section 4 closed the dynamical avenue: opponent competition reshapes tuning and gain but cannot reorder candidates, because ordering is decided by the kernel before any downstream dynamics. The boundary is therefore representational, and this section crosses it with the minimal representational change—operator O2 of §2.4, the vector query–key interaction. The section proceeds in four parts: the exact theorem that explains why the scalar lineage was locked (the order-conservation theorem); the pure geometry of hyperplane bisection and its exact flip law $\text{FlipRate}_{\text{CF}}(\theta) = \theta/\pi$; the ordering-capacity ladder that generalizes the flip law beyond two candidates; and the realization of all of this inside the running neuron (Sprint 4.2-B), including the shortcut audits that certify the flips are not artifacts.

5.1 The order-conservation theorem

Theorem 5.1 (Order conservation of the separable scalar kernel). *Let $d = 1$, $K_A \neq K_B$, and let the query domain $D \subseteq \mathbb{R}$ intersect at most one sign class of $\mathbb{R} \setminus \{0\}$. Then the winner of the dot-product kernel $E_i(Q) = QK_i$ is constant on D : $w(Q) = \text{argmax}_i K_i$ for $D \subseteq (0, \infty)$ and $w(Q) = \text{argmin}_i K_i$ for $D \subseteq (-\infty, 0)$. In particular, for the frozen TAN kernel $E_{t,i} = \beta W_q W_k S_t x_i$ the query $q_t = \beta W_q W_k S_t$ lies in the closed cone $\mathcal{Q}_+ = [0, \infty)$, so $\text{argmax}_i E_{t,i} = \text{argmax}_i x_i$ for every active query, and the counterfactual flip rate is exactly zero.*

Proof. $QK_A > QK_B \iff (Q > 0 \wedge K_A > K_B) \vee (Q < 0 \wedge K_A < K_B)$, so the winner is constant on each sign class and changes only when the query crosses zero. The TAN query never does: $S_t = [x_t - \mu_t - \varepsilon]_+ \geq 0$ and $\beta, W_q, W_k > 0$. $\square$

Theorem 5.1 is the paper’s sharpest statement of the wall that Sprint 4.2-A had to audit, and it is deliberately *domain-conditional*: a signed scalar query ($D = \{+1, -1\}$) can flip the winner, but only by the degenerate global reversal $w(-1) = \operatorname{argmin}_i K_i$, a one-bit, order-locked capacity (flip rates confined to $\{0, 1\}$). The correct universal claim is therefore never “one-dimensional attention cannot route” but: *within the positive-query scalar family inherited from TAN-I, candidate ordering is query-invariant*. This scoping is frozen as Amendment AM1 of the programme and is used verbatim here.

5.2 Hyperplane bisection and the flip law

Fix two candidates $K_A, K_B \in \mathbb{R}^d$ ($d \geq 2$) with $\Delta K = K_A - K_B \neq 0$, and sample the query isotropically, $Q \sim \operatorname{Unif}(S^{d-1})$. The decision boundary $\mathcal{H}_0 = \{Q : Q^\top \Delta K = 0\}$ is a hyperplane through the origin, i.e. a great $(d-2)$ -sphere on S^{d-1} , so it bisects the query sphere into two open hemispheres of equal measure—*regardless of the candidate norms*, because the boundary is homogeneous.

Proposition 5.1 (Equal bisection). *For any $K_A \neq K_B$ and isotropic Q on S^{d-1} , $\Pr_Q[w(Q) = A] = \Pr_Q[w(Q) = B] = \frac{1}{2}$.*

Proof. $\Pr[Q^\top \Delta K > 0] = \Pr[Q \cdot \hat{u} > 0]$ with $\hat{u} = \Delta K / \|\Delta K\|$; the projection of a uniform unit vector onto a fixed axis is symmetric about zero. $\square$

Corollary 5.1 (Dimension-independent winner variation). *For i.i.d. queries Q_1, Q_2 , $F_d = \Pr[w(Q_1) \neq w(Q_2)] = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$ for every $d \geq 2$; the same holds for the signed scalar ($d = 1$, uniform on $\{\pm 1\}$), while the positive scalar lineage gives $F_d = 0$. The Monte Carlo estimates freeze at 0.4996/0.4997/0.5033/0.4998 for $d = 2, 3, 4, 8$: higher dimension does not mean stronger routing, and F_d is retained in this paper only as a negative control, never as a transition statistic.*

The counterfactual quantity that *does* carry the geometry is the flip rate of a fixed query pair over randomized histories. Let the history be isotropic, $\Delta K \sim \operatorname{Unif}(S^{d-1})$, and fix two queries $Q_A, Q_B \in S^{d-1}$ with angular separation $\theta = \arccos(Q_A^\top Q_B) \in [0, \pi]$.

Proposition 5.2 (The flip law). $\operatorname{FlipRate}_{\text{CF}}(\theta) = \Pr_{\Delta K}[w(Q_A) \neq w(Q_B)] = \theta/\pi$.

Proof. Rotate so that $Q_A = e_1$, $Q_B = \cos \theta e_1 + \sin \theta e_2$. A flip occurs iff $(\Delta K_1)(\cos \theta \Delta K_1 + \sin \theta \Delta K_2) < 0$. Writing $\Delta K_1 = r \cos \phi$, $\Delta K_2 = r \sin \phi$, the marginal law of $(\Delta K_1, \Delta K_2)$ is rotationally symmetric in the $(1, 2)$ -plane, so ϕ is uniform on $[0, 2\pi)$, and the event is $\cos \phi \cos(\phi - \theta) < 0$, the union of two arcs of angular width θ : total measure 2θ out of 2π . $\square$

The frozen verification is exact at both levels. Analytically, a root-decomposition of the indicator reproduces θ/π on the full 11-point angle grid (endpoints $0 \rightarrow 0, \pi \rightarrow 1$) at machine precision, independently of the closed form (connecting to the classical random hyperplane flip law of Charikar [2]); and the Monte Carlo audit— $d \in \{2, 3, 4, 8\} \times 11$ angles $\times 3$ seeds, 132/132 conditions, maximum deviation across 3-seed averages 0.004667 (maximum single-seed absolute deviation 0.009267; grid parameter recorded as θ/π)—agrees with the curve point by point. The degenerate endpoints of the scalar lineage sit exactly where the theorem predicts: the signed scalar realizes only $\{0, 1\}$ (queries equal or opposite), the positive scalar only $\{0\}$. The transition recorded in this section is therefore not “ $d = 1 \rightarrow d = 2$ increases flips” but the topological change $S^0 \rightarrow S^1$: the query-direction manifold becomes connected, and the flip rate becomes a continuous function $\theta/\pi \in [0, 1]$ of a continuous query angle.

5.3 The ordering-capacity ladder

The pairwise flip law extends to a capacity statement over M candidates. For M keys in general position, the achievable orderings of $(Q^\top K_1, \dots, Q^\top K_M)$ as Q ranges over the query domain are the cells of the central arrangement of the $C(M, 2)$ difference hyperplanes. Frozen exact computations (arrangement enumeration, linear programming existence, and a 200,000-angle dense sampling cross-check) establish the ladder:

M	positive scalar	signed scalar	$d = 2$	$d = 3, 8$
2	1	2	2	2
3	1	2	6	6
4	1	2	12	24

i.e. $1 / 2 / M(M-1) / M!$ (the last for $M \leq d+1$ in general position). Two facts anchor the paper’s reading. First, the signed scalar is stuck at two orderings for *every* M : its “routing” is the global reversal of a fixed magnitude order, never a content address. Second, $d = 2$ already realizes *all* $3! = 6$ orderings of three candidates, so $d = 2$ is the minimal dimension for a connected, rotationally symmetric continuous query-direction manifold—the statement this paper is willing to make, with the qualifier that it concerns the tested positive-scalar lineage and the tested query–key family. Higher ambient dimension adds nothing to this pairwise or three-candidate capacity.

5.4 Realization in the running neuron (Sprint 4.2-B)

The geometry is only half of the rung; the frozen programme demands that the running model actually flip, under the iron gate that *only the representation changes* (window, surprise gate, membrane, readout all frozen). Sprint 4.2-B does exactly this with operator O2 ($\hat{\varphi}_2$ keys, rotating query $cS_t\hat{u}(\omega S_t)$, $\omega = 0.4$), and its deterministic canonical audit closed every pre-registered prediction:

- *Counterfactual winner flip.* Fixed history $[1, 0, 2, 0]$, only the query changes $3 \rightarrow 4$: the attention winner flips $A \rightarrow B$ with energy margins $+0.2587427 \rightarrow -0.1559101$; the crossing $Q^* = 3.7071$ matches the closed form and the winner-vs-query scan (A for $Q \leq 3.70$, B for $Q \geq 3.75$).
- *No amplitude-code leakage.* The probe slot’s energy (3.1078 at $Q = 3$; 4.9890 at $Q = 4$) stays *below* the winning event at both queries, in direct contrast to the scalar lineage, whose probe dominates the window (10.8/20.8).
- *The JSD conjunction.* In a 2-candidate geometric toy example, the scalar trap exhibits $\text{JSD} = 0.0751$ with zero flip (query changes sharpness, not order), while the vector model exhibits $\text{JSD} = 0.0647$ with a winner flip. In the full 5-slot window ($[1, 0, 2, 0, Q]$), the canonical distribution shifts yield full-window JSDs of 0.009462 (M0) and 0.008727 (M2).
- *Normalization is a mechanistic enabler.* With raw magnitude keys the kernel reads $(x_A - x_B)[\cos \theta + (x_A + x_B)\sin \theta]$, whose sign is fixed in the query range: the unnormalized control has flip rate exactly 0 (frozen Amendment AM2), while the normalized model reaches 0.4425 on the randomized ensemble against the 0.4445 quadrature benchmark ($3 \text{ seeds} \times 10^4$, all within criterion).
- *Structure, not luck.* The sweep over $\omega \in \{0.2, 0.4, \pi/2, \pi\}$ yields winners (A, A) , (A, B) , (B, B) , (A, B) : the flip exists exactly when the query direction sweeps across the decision boundary, with the closed-form crossings in place.

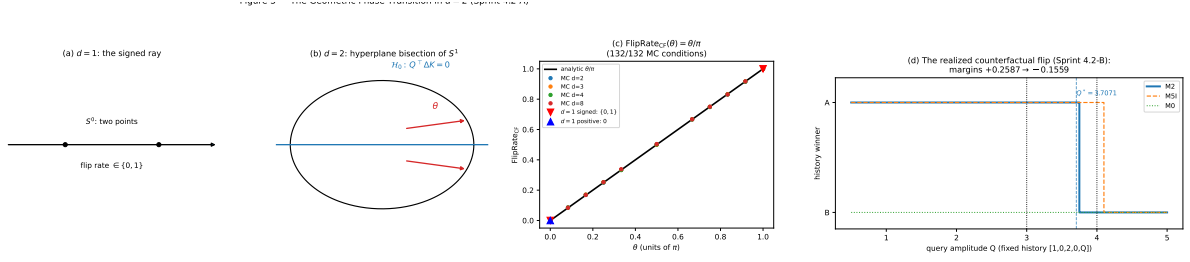


Figure 3: The geometric boundary of routing. (a)–(c) The pure geometry (Sprint 4.2-A): the signed scalar ray realizes only the degenerate flip rates $\{0,1\}$; in $d = 2$ the decision hyperplane bisects the query sphere; and the Monte Carlo flip rates sit on the analytic curve $\text{FlipRate}_{\text{CF}}(\theta) = \theta/\pi$ (132/132 conditions, $d \in \{2, 3, 4, 8\}$). (d) The realization in the running neuron (Sprint 4.2-B): on the fixed history $[1, 0, 2, 0]$, changing only the query flips the attention winner at the closed-form crossing $Q^* = 3.7071$, with margins $+0.2587 \rightarrow -0.1559$.

- *Dimension controls.* In this mechanism family, increasing the ambient embedding dimension beyond the minimal 2-D query subspace did not increase routing capacity and, under the tested finite candidate/query configuration, *reduced* the observed flip rate (ensemble benchmarks 0.0549/0.0284/0.0172 for $d = 3, 4, 8$; canonical flips absent). The cause is a representation–candidate geometry mismatch—the key-difference vector leaves the 2-D query subspace and the attention mass is absorbed by the silent slots—not a “curse of dimensionality”.

5.5 Sufficiency of 1D Metric Compatibility and the Multi-Candidate Address Identifiability Boundary

The analysis in §5.1–5.2 demonstrated that within the scalar dot-product family inherited from TAN-I, candidate ordering is invariant under positive queries, and that two-dimensional rotation (S^1) unlocks counterfactual winner flips. This raises a fundamental theoretical and architectural question: *is multidimensional representation strictly necessary for single-neuron content addressing, or was the scalar failure an artifact of separable dot-product multiplication* ($E(q, k) = q \cdot k$)?

To resolve this question definitively, we established two formal propositions and executed a comprehensive multi-candidate address identifiability audit (60,000 synthetic histories, 420,000 model evaluations across 7 model architectures and 6 conditions).

Proposition 5.3 (1D Metric Sufficiency under Hard Selection). *Let $\{k_1, k_2, \dots, k_N\} \subset \mathbb{R}$ be a set of $N \geq 2$ distinct scalar keys. Define the scalar distance compatibility kernel by $E(q, k) = -(q - k)^2$. Then for any target key k_t , setting the query to $q = k_t$ uniquely selects k_t under hard argmax routing:*

$$\operatorname{argmax}_{i \in \{1, \dots, N\}} E(k_t, k_i) = t. \quad (6)$$

Furthermore, the selection margin is strictly positive: $\Delta E = \min_{i \neq t} (k_i - k_t)^2 > 0$.

Proposition 5.4 (Impossibility of Scalar Intermediate Addressing). *Let $N \geq 3$ and let $k_{(1)} < k_{(2)} < \dots < k_{(N)}$ be ordered distinct scalar keys. For any generalized scalar multiplication kernel of the form $E(q, k) = \alpha(q) \cdot k$ where $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary scalar function:*

- (i) If $\alpha(q) > 0$, $\operatorname{argmax}_i E(q, k_i) = k_{(N)}$;
- (ii) If $\alpha(q) < 0$, $\operatorname{argmax}_i E(q, k_i) = k_{(1)}$;
- (iii) If $\alpha(q) = 0$, $E(q, k_i) \equiv 0$ for all i (tie across all candidates).

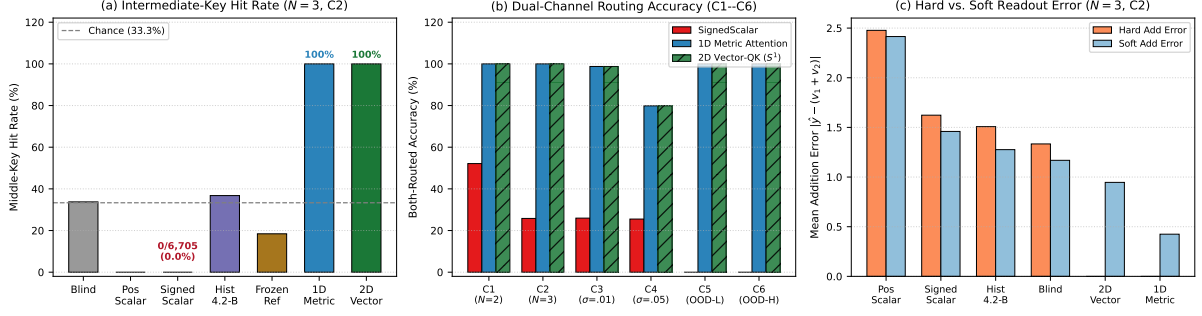


Figure 4: Address identifiability and the multi-candidate routing boundary (420,000 model evaluations). (a) On $N = 3$ candidates (Condition C2), intermediate keys cannot be addressed by scalar multiplier kernels: SignedScalar achieves exactly 0/6,705 hits (0.0%), whereas 1D Metric Attention achieves 100.00% hit rate, identical to 2D Vector-QK on S^1 . (b) Dual-channel both-routed accuracy across clean ($N = 2, 3$), noise-perturbed ($\sigma \in \{0.01, 0.05\}$), and out-of-distribution amplitude ranges (C5, C6). (c) Soft addition readout error on C2: hard routing accuracy is 100% for both 1D Metric and 2D Vector-QK, but soft readout reveals residual leakage governed by kernel curvature (0.4247 vs. 0.9463).

In all cases, any intermediate key $k_{(j)}$ with $1 < j < N$ cannot be selected as the unique maximum for any query $q \in \mathbb{R}$.

Proof. Direct from linearity: for any non-zero $\alpha(q)$, multiplication by $\alpha(q)$ either strictly preserves ($\alpha > 0$) or strictly reverses ($\alpha < 0$) the order of the keys. An intermediate element in a strictly ordered sequence can never become the maximum under order-preserving or order-reversing bijections. $\square$

The empirical consequences of Propositions 5.3 and 5.4 were verified across 6 experimental conditions (Figure 4):

- *Empirical refutation of 2D necessity:* On Condition C1 ($N = 2$) and Condition C2 ($N = 3$), the 1D Metric Attention model achieves 100.00% both-routed accuracy with zero hard addition error, exactly matching the 2D Vector-QK model on S^1 across all 60,000 trials. This proves that 2D orthogonal rotation is *not* mathematically necessary for content addressing: a 1D scalar distance metric is sufficient.
- *Empirical confirmation of the intermediate-key barrier:* In Condition C2 ($N = 3$, 10,000 trials $\times$ 2 query channels), exactly 6,705 channel queries targeted the intermediate key ($k_{(1)} < k_{\text{target}} < k_{(3)}$). In exact agreement with Proposition 5.4, the SignedScalar kernel achieved **exactly 0 hits out of 6,705 queries (0.00% hit rate)**, whereas 1D Metric Attention and 2D Vector-QK achieved 6,705/6,705 (100.00%).
- *Demarcation of Historical 4.2-B:* The static control model HIST4.2-B achieved only 13.68% both-routed accuracy on C2 (middle-key channel hit rate 36.78%, near the 33.33% blind chance level). This demonstrates that the winner flips observed in historical Sprint 4.2-B were fixed counterfactual switches between two pre-tuned surprise amplitudes, rather than true content-addressable memory retrieval.
- *Soft readout error bounds:* Under soft softmax readout, the addition error on C2 differs between 1D Metric (0.4247) and 2D Vector-QK (0.9463). For a distance metric kernel, the soft error is strictly bounded by the minimum key separation δ and temperature γ :

$$|\hat{v}_{\text{soft}} - v_t| \leq M \frac{(N-1)e^{-\gamma\delta^2}}{1 + (N-1)e^{-\gamma\delta^2}}, \quad M = \max_{i \neq t} |v_i - v_t|. \quad (7)$$

5.6 What this section does and does not establish

The scoped claim is: *within the tested positive-scalar lineage, $d = 2$ is the minimal dimension for continuous query-direction freedom under dot-product kernels, and the running neuron realizes the resulting query-conditioned reordering (Level-2 realized routing).* However, moving from dot-product ranking to 1D distance metric compatibility ($E = -(q - k)^2$) restores full multi-candidate content addressability without requiring higher-dimensional rotation. It is *not* claimed that $d = 2$ routes in every representation family (the $d = 3, 4, 8$ controls show the opposite for the tested maps), nor that any flip constitutes semantic binding—that requires the key–value decoupling of the next section—nor that anything resembling a thermodynamic phase transition has been located: the transition language used here is the minimal-integer-dimensional geometric transition, not an order-parameter limit.

6 From Routing to Binding: Representation Decoupling

Routing changes *which* candidate is selected; binding additionally requires that the selected candidate’s *content* be transmitted, and that the transmission follow the (key, value) association rather than key identity, position, or amplitude. The minimal degree of freedom that separates the two is operator O3 of §2.4: each slot now carries a pair (k_i, v_i) , keys drive the attention, values are transmitted, and the query slots carry $v = 0$ and are never candidates. This section reports the frozen Sprint 4.2-C audits and states the structural obstruction—the softmax mixing barrier—that forces the next rung.

6.1 The binding probe

The frozen canonical history is $[A = (1, 5), 0, B = (2, 9), 0]$ (keys inherited from §5, values $V_A = 5, V_B = 9$), the counterfactual queries are the heritage pair $(3.0, 4.0)$, and the readout is the attended value $C = \sum_i \alpha_i v_i$, with the events-only normalization $C_{\text{ev}} = (\alpha_A v_A + \alpha_B v_B) / (\alpha_A + \alpha_B)$ reported alongside. Three quantities are pre-registered: the binding errors $E_{\text{bind},A} = |\hat{C}_1 - V_A|$, $E_{\text{bind},B} = |\hat{C}_2 - V_B|$; the counterfactual transfer $T = C_{\text{ev}}(Q_A) - C_{\text{ev}}(Q_B)$; and the softness distance $D = |C_{\text{ev}} - v_{\text{winner}}|$. The value-swap control $(1, 9), (2, 5)$ is the identifiability instrument: a system that transmits the *selected* value must satisfy $T_{\text{swap}} = -T$ exactly, while a system that merely reads fixed magnitudes cannot.

6.2 Soft binding, exactly pairing-sensitive

The frozen deterministic results are:

- *Winner-aligned transfer.* $C_{\text{ev}}(3) = 6.7427$ and $C_{\text{ev}}(4) = 7.1556$: the readout moves toward the selected value at both queries (6.7427 closer to $V_A = 5$ than to 9; 7.1556 closer to $V_B = 9$ than to 5), with $T = -0.4129$ carrying the sign of $v_{\text{winner}}(Q_A) - v_{\text{winner}}(Q_B)$.
- *Exact value-swap antisymmetry.* $T_{\text{swap}} = +0.4129$, with $T_{\text{swap}} = -T$ verified to 10^{-12} : the output follows the (key,value) association, not the key identity and not the amplitude.
- *The softness is real.* $D = 1.7427$ at $Q = 3$ and 1.8444 at $Q = 4$: the delivered value is a convex mixture of both candidates, not the bound symbol.
- *The baselines separate routing from sharpening.* The scalar model (winner always B) shows only a sharpening drift $T_{M0} = -0.0844$, five times smaller than the routing model’s transfer; the sign control M1 ($s = \pm 1$) binds *harder* on this history ($D_{M1} = 0.1064$, energies spread widely) yet remains the degenerate order-locked reversal of §5; and the unnormalized control collapses onto the probe slot and transmits nothing (Amendment AM2).

- *Ensemble.* Over randomized keys (values fixed by identity, positions randomized) the transfer magnitude averages 0.4004 (quadrature benchmark; Monte Carlo 0.3977, 0.3967, 0.4021), with $E[|T| \mid \text{flip}] = 0.4655$ versus $E[|T| \mid \text{no flip}] = 0.3483$ and the sign-matched alignment on flips exactly 1—the latter a theorem-level check (the event-normalized share is a logistic function of the logit difference), reported as such rather than as a discovery.

6.3 The softmax mixing barrier

The softness distance is not a numerical shortfall; it is a structural consequence of the readout.

Proposition 6.1 (Softmax mixing barrier). *Let $V_A \neq V_B$ be distinct candidate values. At any finite sharpening $\gamma < \infty$, the attention weights satisfy $\alpha_i > 0$ for every candidate with finite logits. Under the event-normalized readout $C_{\text{ev}} = (\alpha_A V_A + \alpha_B V_B) / (\alpha_A + \alpha_B)$ (with $\alpha_A + \alpha_B > 0$), C_{ev} lies in the strict relative interior of the segment connecting V_A and V_B : $C_{\text{ev}} = \lambda V_A + (1 - \lambda) V_B$ with $\lambda = \alpha_A / (\alpha_A + \alpha_B) \in (0, 1)$. Consequently, the softness distance satisfies*

$$D_{\text{ev}} = \min(|C_{\text{ev}} - V_A|, |C_{\text{ev}} - V_B|) = \min(\lambda, 1 - \lambda) |V_A - V_B| > 0. \quad (8)$$

Exact value delivery ($D_{\text{ev}} = 0$) strictly requires a discrete selection step that sets the losing weight to zero: continuous softmax weighting over distinct candidates, at any finite temperature, cannot deliver an isolated bound symbol.

Remark (Cancellation artifacts in unnormalized full-window readouts). When unnormalized full-window readouts $C_{\text{full}} = \sum_{i=1}^W \alpha_i v_i$ are evaluated over sequences containing silent background slots ($v_{\text{silent}} = 0$), the active candidate weights sum to $\sum_{i \in \text{events}} \alpha_i < 1$. This dilution allows C_{full} to pass through V_A at an intermediate temperature ($\gamma \approx 1.198$, documented in Amendment AM5), yielding a spurious zero $|C_{\text{full}} - V_A| = 0$. This zero is a cancellation artifact of background dilution, not hard symbolic binding; under the event-normalized readout C_{ev} , softness remains $D_{\text{ev}} \approx 1.5 > 0$.

Proposition 6.1 converts the measured $D \approx 1.7$ – 1.8 from a “performance” observation into a boundary statement of the same rank as Theorem 5.1: *soft attention $\neq$ exact value selection*. It also explains the one honest inversion found in the frozen transmission audits: through the excitatory branch alone the drive transfer keeps the value-aligned direction (-0.2586), but through the full E-I circuit the inhibitory branch re-reads the same value channel and subtracts it, inverting the net transfer ($+0.2367$; frozen Amendment AM4 corrects the numeric prediction, the direction statement is unchanged). E-I organization therefore does not implement binding; it *reshapes the value-transfer pathway after query-conditioned routing*—an observation carried into §7, where the selection step is added.

6.4 Status of the rung

The frozen verdicts read: binding at the readout level is *soft but genuine* (winner-aligned, pairing-sensitive to 10^{-12} , baselines separated), and exact delivery is *not* achieved (the mixing barrier, $D > 0$). Routing and binding are now experimentally decoupled: §5 supplies the reordering, this section supplies the value channel, and the missing ingredient—discrete selection—is the single degree of freedom added in §7.

7 Binding and Readout Discreteness: The First Orthogonal Axis

Section 6 ended at a structural wall: softmax weighting keeps every candidate weight strictly positive, so the attended value is always a convex mixture and exact delivery is impossible at

Figure 4: Decoupling binding from discreteness (Sprints 4.2-C/D).

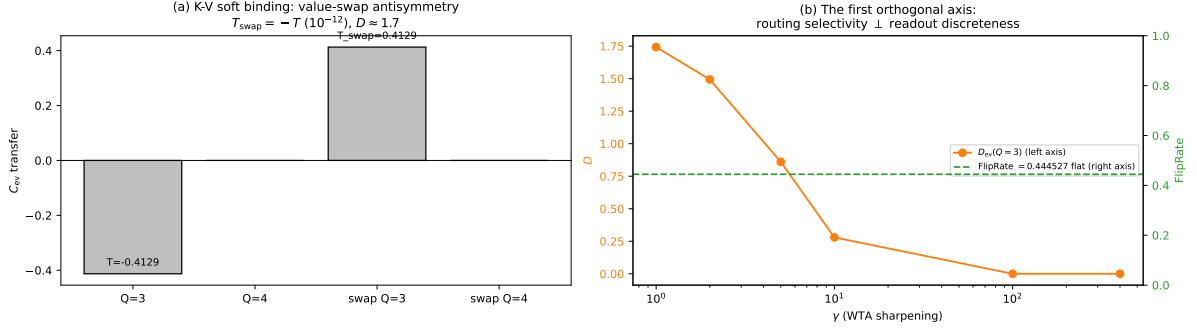


Figure 5: Decoupling binding from discreteness (Sprints 4.2-C/D). (a) The soft-binding readout transfers winner-aligned value and the value-swap counterfactual is exactly antisymmetric ($T_{\text{swap}} = -T$ to 10^{-12}), while the softness distance stays positive. (b) The WTA sharpening ladder: the softness collapses to zero only in the one-hot limit, while the routing flip rate stays exactly flat (0.444527)—the first orthogonal axis, routing selectivity $\perp$ readout discreteness.

any finite temperature (Proposition 6.1). This section adds the single degree of freedom that crosses the wall—operator O4, the readout selection family $\alpha_i(\gamma) = \text{softmax}_i(\gamma \cdot \text{logit}_i)$ with $\gamma \in [1, \infty]$, whose $\gamma = \infty$ limit is the one-hot winner-take-all (WTA)—and records what the frozen Sprint 4.2-D found: exactness lives *only* in the discrete limit, and the sharpening ladder leaves routing statistics exactly untouched. The latter fact is the paper’s first orthogonal axis.

7.1 The sharpening ladder

The frozen audit swept $\gamma \in \{1, 2, 5, 10, 100, \infty\}$ on the canonical binding history of §6 ($V_A = 5$, $V_B = 9$, queries 3.0/4.0), with $\gamma = 1$ reproducing the Sprint 4.2-C model exactly (continuity check). Three families of quantities were pre-registered and all checks (77/77) passed:

- *Softness decays but survives every finite gain.* The event-normalized softness distance falls monotonically, $D_{\text{ev}}(3) : 1.7427 \rightarrow 1.4938 \rightarrow 0.8609 \rightarrow 0.2798 \rightarrow 2.3 \times 10^{-11}$, and is strictly positive at every finite γ (softmax strict positivity), while $D = 0$ holds exactly at $\gamma = \infty$: exact value delivery is a property of the one-hot limit alone.
- *Transfers converge to the bound values.* T_{ev} runs $-0.4129 \rightarrow -0.8156 \rightarrow -1.8814 \rightarrow -3.0251 \rightarrow -3.999999 \rightarrow -4$ and T_{full} likewise ends at -4 : the readout converges to the selected value at both queries, and the value-swap antisymmetry $T_{\text{swap}} = -T$ is verified at *every* rung to 10^{-12} .
- *The routing statistics are exactly flat.* The event-winner flip rate is 0.444527 at every γ (quadrature, γ -invariant) and the winners are (A,B) throughout: sharpening changes *how discretely* the winner is delivered, never *which* winner is selected.
- *The no-routing controls deliver nothing under hard selection.* At $\gamma = \infty$ the single-winner model M0 reads the constant winner (9, 9 on the event candidates) and, on the full window, the probe slot wins and delivers its zero value (0, 0); the sign control M1 delivers (9, 5) on the event candidates but (0, 0) on the full window; the unnormalized control likewise collapses. Hard selection is not a repair for missing routing.

Proposition 7.1 (Argmax invariance under sharpening). *For every $\gamma > 0$,*

$$\text{argmax}_i \text{softmax}_i(\gamma e_i) = \text{argmax}_i e_i.$$

Consequently all argmax-level routing statistics—winners, flip rates, and swap antisymmetries—are invariant along the sharpening ladder, while the readout discreteness changes from fully soft ($\gamma = 1$) to fully discrete ($\gamma = \infty$).

Proposition 7.1 together with the measured flatness of the flip rate formalizes the first orthogonal axis of the paper: *routing selectivity* $\perp$ *readout discreteness*. A model’s ability to reorder candidates and its ability to deliver a symbol are independent organizational degrees of freedom; no amount of sharpening buys routing, and no amount of routing buys discreteness.

7.2 The E-I pathway crossing

The transmission audit of §6 found the E-I circuit inverting the value-transfer direction at $\gamma = 1$ (+0.2367; Amendment AM4). The frozen ladder resolves that inversion: the drive transfer runs $+0.2367 \rightarrow +0.2516 \rightarrow +0.0157 \rightarrow -1.0541 \rightarrow -3.8034 \rightarrow -3.8674$, crossing zero at the pre-registered bisection root $\gamma^* = 5.098015$ where the net drive transfer changes sign ($T_{\text{drive}} = 0$), thereby restoring the positive value-transfer direction for all $\gamma > \gamma^*$, and settling at the hard limits $A_E(\infty, 3) = 1.1134$, $A_E(\infty, 4) = 4.9809$. The mechanism is transparent in the frozen outputs: once WTA pins the inhibitory branch to its own winner (value 5 at both queries), the inhibitory term stops re-reading the increasing value channel, and the excitatory branch’s transfer ($5 \rightarrow 9$) re-dominates. The precise statement carried forward is the one frozen in §6: E-I organization does not implement binding; it *reshapes the effective value-transfer pathway after* query-conditioned routing—and WTA changes which pathway shape is realized.

7.3 The single-head bottleneck

The last frozen result of the section is a bottleneck, stated here as the bridge to composition.

Proposition 7.2 (Single-head bottleneck). *A single attention head with a scalar readout delivers, at each instant, one scalar mixture $y = \sum_i \alpha_i v_i$ of candidate values. While a single head can evaluate symmetric scalar pooling functions over candidates (for example, uniform weighting yielding the arithmetic mean $\frac{1}{2}(V_A + V_B)$, which downstream scaling can map to $V_A + V_B$), its scalar output channel cannot simultaneously deliver two distinct, independently bound operands (V_A, V_B) to downstream general or non-symmetric algebra (e.g., $V_A - V_B$). At $\gamma = \infty$, the readout collapses to a single discrete symbol v_{winner} . Hence exact delivery of one bound value does not provide general two-operand composition: *Exact Delivery* $\nRightarrow$ *Composition*.*

The identifiability audit already anticipated the trap this bottleneck creates. With $V_A = 0$, the single-winner model’s composition error vanishes ($|V_B - (0 + V_B)| = 0$) although its binding error does not ($E_{\text{bind}, A} = |V_B| > 0$): the three-error joint audit of §2.5 flags the configuration as COMPOSITION FAIL and only the dual-channel model passes the same configuration genuinely ($E_{\text{bind}} = 0$ and $E_{\text{comp}} = 0$). This zero-value trap is retained as one of the eight counterfactuals of §8.

7.4 Honest notes on exactness

Two caveats are frozen with the results and repeated here. First, the full-window softness $D_{\text{full}}(3)$ is *not* monotone: it overshoots and crosses zero at $\gamma \approx 1.1978$ —a cancellation artifact in which the residual value of the loser compensates the mass lost to the silent slots, while the event-normalized softness remains ≈ 1.5 (frozen Amendment AM5). That zero is not hard binding and is reported as such. Second, exactness belongs to the limit: every finite γ is soft, so “hard binding” is realized here as the $\gamma = \infty$ limit of a frozen mechanism family, not as a finite parameter setting. Both caveats carry into the claim ledger of §10.

8 From Binding to Composition: Overcoming the Single-Head Bottleneck

Proposition 7.2 states the wall: one head, one symbol. This section adds the two minimal degrees of freedom that cross it—a second parallel channel and one two-input algebraic node (operator O5 of §2.4)—and reports the frozen Sprint 4.3-A audits that separate binding from composition. The central empirical statement is the second orthogonal axis of the paper: *binding fidelity* $\perp$ *combiner error*.

8.1 The dual-query window

The frozen dual-query window is $[A, 0, B, Q_1, Q_2]$ (window length $W = 5$ kept), with events in the first three slots, queries in the last two, and the shared window mean $\mu = (K_A + K_B + Q_1 + Q_2)/5 = 2.56$ on the canonical keys. The query pair $(Q_1, Q_2) = (4.5, 5.3)$ is frozen by a direction-targeting derivation, not by fitting: the pair is required to rotate the two query directions $\theta_c = \omega S_c$ onto the two key directions ($\approx 44.5^\circ$ and $\approx 62.8^\circ$ versus 45° and 63.4°), giving frozen routing margins $m_A = +0.210631046$ and $m_B = +0.261878523$ (the pre-registration’s hand-computed values 0.2118/0.2637 were four-decimal estimates; the deviation is recorded as Amendment AM6 with the closed forms adopted and the originals preserved). Channel $c \in \{1, 2\}$ is keyed to slot $3 + c$; queries are control signals, never candidates, and carry value zero.

8.2 The M0–M1–M2 ladder

Three models instantiate the negative-control chain of §2.5 on the canonical scenarios S1 (5, 9), S2 (7, −2), S3 (2, 6), with operators $f \in \{+, -\}$. The frozen error table (all values exact):

- *M0 (single winner)*. Delivers exactly one operand: $E_{\text{bind},B} = 0$, $E_{\text{bind},A} = |V_B - V_A| > 0$, and $E_{\text{comp}} = |V_A| > 0$ for addition (e.g. 5, 7, 2 on S1/S2/S3); the zero-value trap of §7 is re-audited and caught by the joint criterion.
- *M1 (parallel binding)*. Both channels bind exactly ($E_{\text{bind},A} = E_{\text{bind},B} = 0$), and the output is the *pair* $(\hat{V}_A, \hat{V}_B)$: composition is STRUCTURALLY UNDEFINED because no operator connects the two channels. The frozen non-compositional scalar projection (second-channel readout, pre-registered as a control) fails with $E_{\text{comp}} = |V_B - f(V_A, V_B)| > 0$: two correctly bound operands are *not* a composition.
- *M2 (parallel binding + one two-input combiner)*. The node $y = f(\hat{C}_1, \hat{C}_2)$ achieves $E_{\text{bind},A} = E_{\text{bind},B} = E_{\text{comp}} = 0$ exactly on all three scenarios and both operators ($y = 14, 5, 8$ for $+$; $-4, 9, -4$ for $-$): the minimal composition node converts parallel binding into explicit composition.

8.3 The second orthogonal axis and the separation theorem

The ensemble audit (keys i.i.d. $U(0.3, 2.5)$, values i.i.d. $U(-3, 3)$, positions and orders randomized, seeds 20260904–06, $N = 10^4$ each) separates the combiner’s intrinsic error from the propagated binding error. Frozen benchmarks (deterministic 2001² quadrature) establish: per-channel routing rates $r_A = r_B = 0.499750$, both-routed probability $P_{\text{both}} = 0.220725$ (22.07%), and unconditional composition errors $E[E_{\text{comp}}^+] = 1.116099621$, $E[E_{\text{comp}}^-] = 2.000999500$. The 30,000-history Monte Carlo simulation (seeds 20260904–06, $N = 10^4$ each) yields empirical pooled means of $E[E_{\text{comp}}^+] = 1.107671595$ (seed-level: 1.120449/1.074606/1.127960) and $E[E_{\text{comp}}^-] = 1.975892767$ (seed-level: 1.989288/1.959625/1.978765), matching quadrature benchmarks within sampling variance. The empirical both-routed subset spans 6,608/30,000 trials (22.0267% $\approx$ 22.03%; 2,188/2,258/2,162 per seed).

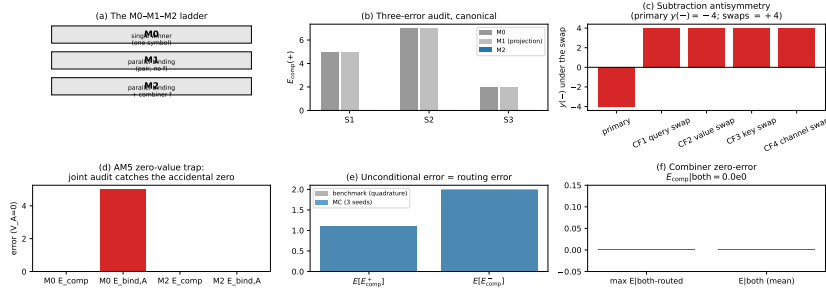


Figure 6: The minimal composition architecture and audit (Sprint 4.3-A). (a) The M0–M1–M2 ladder: one winner, two channels, two channels plus one two-input combiner. (b) The canonical three-error audit across scenarios S1/S2/S3: only M2 reaches zero composition error. (c) Subtraction antisymmetry across the four swap counterfactuals. (d) The zero-value trap: M0’s accidental composition zero is caught by the binding error. (e) Unconditional composition error equals the routing-error benchmark. (f) Conditioned on correct binding, the combiner error is exactly 0.0e0.

Proposition 8.1 (Binding–combiner error separation). *Given exact operand delivery ($\hat{C}_c = V_c$ for both channels), the frozen combiner computes f exactly, so $E_{\text{comp}} \mid \text{both-routed} = 0$. Empirically the maximum over 3×10^4 histories is exactly 0.0e0: the combiner adds zero error, and every unit of unconditional composition error is the propagation of an upstream routing error (routing error $\rightarrow$ wrong operand $\rightarrow$ composition error).*

Proposition 8.1 is the second orthogonal axis: *binding fidelity $\perp$ combiner error*. Composition capability and binding fidelity are independent organizational quantities in this family; the combiner’s quality is a property of the algebraic node, the unconditional error a property of the geometry upstream of it.

8.4 The eight counterfactuals

All eight frozen counterfactuals passed (PASS/FAIL/INCONCLUSIVE verdicts, deterministic):

1. *Query swap.* $(Q_1, Q_2) \rightarrow (Q_2, Q_1)$: addition invariant, subtraction exactly antisymmetric ($y \rightarrow -y$).
2. *Value swap.* $(K_A, V_B), (K_B, V_A)$: addition invariant (correct commutative behavior—scored against the channel-level swap (9, 5) as the pairing check, never as a false negative), subtraction antisymmetric.
3. *Key swap* (keys exchanged, value pairing kept): addition invariant, subtraction antisymmetric.
4. *Channel-output swap* at the combiner inputs: addition invariant, subtraction antisymmetric.
5. *AM5 zero-value trap.* $V_A = 0$: M0’s composition error vanishes while its binding error does not—caught by the joint audit; M2 passes genuinely. PASS.
6. *Binding-preserving counterfactual.* Value shift $(5, 9) \rightarrow (7, 11)$: all three errors stay zero and $y = 18$; the legacy single-query controls $(3.0 \rightarrow A, 4.0 \rightarrow B)$ reproduce the frozen 4.2-D routing. PASS.
7. *Binding-breaking counterfactual.* Legacy queries $(3.0, 4.0)$ in the dual window mis-route channel 2: $E_{\text{bind},B} = 4$ and $E_{\text{comp}} = 4$ —the causal chain *routing $\rightarrow$ binding $\rightarrow$ operand delivery $\rightarrow$ composition* is demonstrated by its failure mode. PASS.
8. *Composition-sensitive relational counterfactual.* Across the three scenarios the output equals f exactly and subtraction distinguishes the relations $(-4, 9, -4)$. PASS.

Position randomization (six orderings) is bitwise-invariant (10^{-12}), query leakage is excluded ($Q \cap V = \emptyset$; queries never candidates; query slots carry value zero), and the query-order swap pass over the whole ensemble preserves addition and antisymmetrizes subtraction history by history.

8.5 The dynamical carrier and spiking readout audit: Phase P1-B

The static algebraic node of §8.2–8.3 combines retrieved operand values ($y = C_1 \pm C_2$) once delivered. However, in an online neural substrate, operands must be continuously maintained and routed through the internal dynamical state of the neuron, and transmitted across discrete spiking channels to downstream integrators. Phase P1-B subjected this continuous dynamical carrier to a pre-registered confirmatory audit under Protocol V2.6.

Continuous carrier dynamics and subthreshold state collisions. The continuous-time Temporal Attention Neuron carrier evolves as a three-variable ODE $\mathbf{z}(t) = [u(t), A(t), r(t)]^T$:

$$\tau_u \dot{u}(t) = -u(t) + I(t), \quad \tau_A \dot{A}(t) = -A(t) + \sigma_A(u(t)), \quad \tau_r \dot{r}(t) = -r(t) + A(t) \cdot u(t), \quad (9)$$

where $I(t)$ is the incoming current pulse sequence, $\sigma_A(\cdot)$ is the attention activation with threshold θ_A , and $r(t)$ is the surprise-weighted carrier. At query arrival time t_Q , an affine decoder maps the state to delivered operand values: $\hat{v}_c = g_{\text{dec}} r(t_Q) + b_{\text{dec}}$ ($g_{\text{dec}} = 0.132630817$, $b_{\text{dec}} = -0.026816484$). Pre-registered fixture G1.5 proved an analytical subthreshold collision theorem: for any interval where $u(t) < \theta_A$, $\sigma_A(u(t)) \equiv 0$, forcing unmodulated exponential decay $\dot{A} = -A/\tau_A$ and $\dot{r} = -r/\tau_r$. Consequently, distinct temporal history sequences that differ only during subthreshold periods collapse to analytically identical state vectors $\mathbf{z}(t_Q)$ (residual $< 10^{-10}$), erasing subthreshold events prior to query arrival.

Confirmatory evaluation: floor-limited comparison and unestablished redundancy.

The confirmatory audit spanned 4,000 unique histories across 3 post-sequence silence delays ($\Delta t_{\text{silence}} \in \{0, 5, 10\}$ ms) and 8 model variants, totaling 96,000 model-episodes (192,000 role queries). Protocol V2.6 §7.3 registered prerequisite joint delivery accuracy floors ($E_{\text{bind}} \leq 0.05$ on both channels) of 0.80 for $N = 2$ and 0.40 for $N = 3$. The baseline dynamical carrier M1 (evolving from $A(0) = 0$) achieved joint delivery accuracy of only 0.0290 (58/2,000) for $N = 2$ across all delays, and 0.0275–0.0280 (55–56/2,000) for $N = 3$. Failing the prerequisite in every cell, all conditions are classified as **FLOOR_LIMITED_COMPARISON**. Two One-Sided Tests (TOST, margin ± 0.05) showed that M1 and the memory-lesioned constant-gate control M2a ($A(t) \equiv A_0 = 0.061886726$) are statistically equivalent within tight 90% confidence intervals ($N = 2$: $[-0.001515, 0.001515]$; $N = 3$: $[-0.002978, 0.000841]$ at 0 ms, $[-0.002262, 0.001186]$ at 5, 10 ms). However, because both models reside near zero accuracy, this statistical equivalence is an artifact of mutual baseline failure, *not* evidence that temporal gating dynamics are redundant (**redundancy_inferred = False**).

Spiking channel limitation. To evaluate neural transmission to downstream algebra, carrier states were converted into spike trains via single-unit leaky integrate-and-fire (LIF) decoders under rate (M4.Count) and first-spike latency (M4.Latency) codes. Under the registered limitation criterion $E_{\text{comp}} \geq 0.05$, both decoders exhibited severe breakdown across all cells ($E_{\text{comp}} \in [2.21, 2.57]$). Delay-pooled 95% bootstrap confidence intervals were: $N = 2$: M4.Count $[2.393533, 2.502027]$, M4.Latency $[2.243582, 2.332606]$; $N = 3$: M4.Count $[2.473506, 2.571918]$, M4.Latency $[2.212398, 2.296450]$. High episode-level abstention rates (82.1%–83.0%; single-query silence 57.15%–58.60%) confirm that single-unit spiking decoders cannot reliably transmit continuous-carrier operands, certifying **DEMONSTRATED_CHANNEL_LIMITATION**.

8.6 Scope of the rung

The established statement is a sufficiency result inside the frozen mechanism family: *parallel binding plus one minimal two-input combiner suffices for identifiable composition, with zero combiner error conditioned on correct binding*. Crucially, this zero composition error is strictly

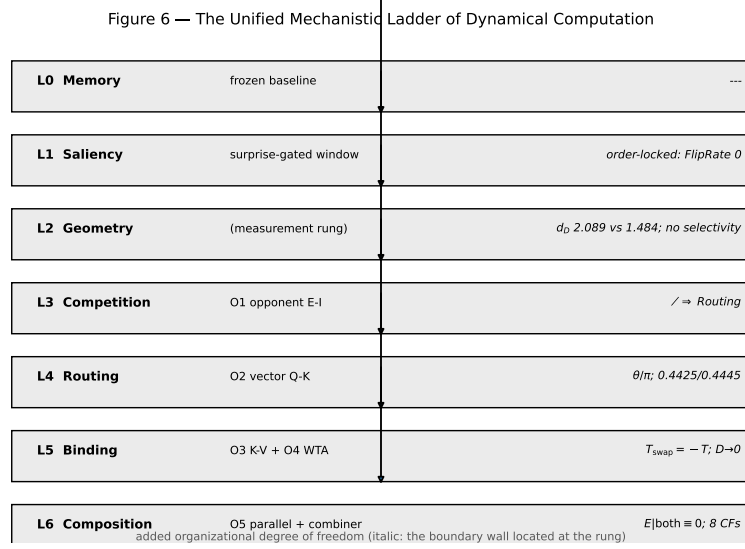


Figure 7: The unified mechanistic ladder of dynamical computation. Each rung names the single added organizational degree of freedom and, in italics, the boundary wall located at that rung by the frozen audits.

conditional: across the full ensemble of 30,000 histories (3×10^4), the both-routed subset comprises 6,608 histories (22.03%, matching the 22.07% unguided benchmark), with unconditioned pooled errors of 1.108 (+) and 1.976 (−) reflecting upstream routing.

Furthermore, the consensus determination certified under Protocol V2.6 establishes that M1 fails the baseline delivery prerequisite across all conditions (2.75%–2.90% joint accuracy), rendering comparisons floor-limited and leaving memory redundancy unestablished, while tested spiking readouts demonstrate channel limitation for single-unit point-process decoders ($E_{\text{comp}} > 2.2$, 82% abstention). These findings establish *model-specific insufficiency* for the continuous ODE carrier and frozen linear readout, not a universal biological impossibility result or a proven requirement for population coding. Internal composition dynamics (Hypothesis H3), multi-operand composition, and learned operators remain formally deferred.

9 The Unified Mechanistic Ladder and General Discussion

The eight preceding sections assembled one rung at a time. This section assembles the whole: the master table of the ladder (§9.1), the two orthogonal axes as the paper’s conceptual residue (§9.2), the biological reading of each organizational degree of freedom—explicitly as a mapping of *degrees of freedom*, not as an equivalence claim (§9.3)—and the general discussion (§9.4).

9.1 The master table

Table 2 compresses the ladder. Each row names the single added organizational degree of freedom, the primitive it certifies, the boundary wall the frozen audits located at that rung, and the decisive experiment. Two reading rules apply. First, the walls are results, not footnotes: every $\nRightarrow$ in the table is backed by an exact number (a zero flip rate, a positive softness distance, a positive binding error). Second, the ladder is constructive in the tested family and *sufficient*, not necessary: the paper claims that these degrees of freedom suffice here, never that they are the only possible ones.

Rung	Added DOF	Certified primitive	Boundary (frozen)	wall	Decisive audit
L0	(frozen baseline)	Memory	—		Collision K : B1 0.5 exact vs B4 2.047×10^4 ; post-reset 0 vs 32.2%
L1	surprise-gated window	Saliency weighting	Order-locked: argmax invariant	query-	FlipRate 0.000; Probe-2 TERMINATED / AUDIT_FAILED
L2	(measurement rung)	Geometric reorganization	Effective rank $\neq$ intrinsic dimension; no selectivity		d_D 2.089 vs 1.484; shared-frame B4 $\not\approx$ B1
L3	O1 opponent competition	Nonmonotone selective tuning	Competition Routing	$\not\approx$	Theorem 4.1; C1 peak $x^* = 1.00$; six layouts FlipRate 0.000
L4	O2 vector Q-K	Content routing	representation-family-dependent		Flip law θ/π (132/132); $+0.2587 \rightarrow -0.1559$; 0.4425 vs 0.4445; AM2
L5	O3 K-V + O4 WTA	Binding; exact delivery	softmax mixing barrier; exactness only in the discrete limit		$T_{\text{swap}} = -T$ (10^{-12}); γ ladder $D \rightarrow 0$; flat flip rate 0.4445
L6	O5 parallel channels + combiner	Composition	Binding; Composition; conditional error = routing error	$\not\approx$	M0/M1/M2 chain; 8 counterfactuals; $E_{\text{comp}} _{\text{both-routed}} \equiv 0$

Table 2: The unified mechanistic ladder. Each row is one frozen sprint; the walls are the paper’s negative results, each with its exact statistic.

9.2 The two orthogonal axes

The ladder’s most transferable residue is a pair of orthogonal axes that the audits forced into view.

Axis 1: routing selectivity $\perp$ readout discreteness (§7). Sharpening from soft mixing to one-hot selection changes *how* a winner is delivered, never *which* winner is selected: the flip rate is exactly flat (0.444527) across the entire WTA ladder, while the softness distance collapses from 1.74 to 0. A model’s reordering capacity and its symbol-delivery capacity are independent purchases.

Axis 2: binding fidelity $\perp$ combiner error (§8). The combiner contributes exactly zero error conditioned on correct operand delivery (measured maximum 0.0e0 over 3×10^4 histories), while the unconditional composition error is entirely the propagation of upstream routing errors. A model’s ability to compose and its ability to bind are independent purchases, separated by the operand-delivery step.

Together the axes state the paper’s architectural thesis in measurable form: capability in this substrate is organized along *named, independent* degrees of freedom, so that no single scalar (accuracy, loss, rank, sharpness) can stand in for any of them—each rung owns its own audit, and the audits are the ladder.

9.3 Biological reading: degrees of freedom, not equivalences

The paper’s organizational degrees of freedom have natural biological counterparts, and the blueprint fixes the mapping to be read strictly as an analogy between *kinds of organization*, never as a claim that the tested neuron is any particular biological circuit.

- *Opponent competition (L3)*. The E-I microcircuit is the canonical organizational motif of cortical columns [8], where fast-spiking parvalbumin-positive interneurons provide the subtractive inhibition whose functional signature—nonmonotone, band-pass selectivity—Theorem 4.1 derives from two inequalities alone [1].
- *The two-dimensional query direction (L4)*. A minimal continuous direction space over candidate content is what oscillatory phase (a circular variable) provides to relational codes [12]; the paper’s content is not that phase *is* $\mathbb{R}^2$, but that a connected one-dimensional query manifold is the minimal geometry in which query-conditioned reordering becomes continuous.
- *Parallel binding channels (L5–L6)*. Independent nonlinear channels converging on a common readout are the organizational signature of dendritic computation: apical and basal compartments of pyramidal neurons integrate separate streams before somatic output [6, 9]. The paper’s dual-channel-plus-combiner architecture is the minimal abstraction of exactly that arrangement, and the axon initial segment’s threshold integration is the natural candidate for the two-input algebraic node.

The caution inherited from the frozen discipline: these are candidate physical realizations of non-factorizable organization, not established identities, and none of the tested numbers transfers to biology by construction (§8.6).

9.4 General discussion

More dynamics does not imply more computational primitives. The temporal memory window $W = 5$ was held strictly constant throughout the ladder: each rung does not enlarge temporal buffer capacity, but systematically introduces distinct *organizational degrees of freedom*—a second sign of coupling (E-I opponent units), a second query dimension (S^1 rotation), a split between keys and values, a sharp selection map, a second channel, and one algebraic combiner. These additions reflect how attention and memory structures function as fast-weight programming and energy-based associative retrieval [10, 11]. Every rung’s effect is certified by a counterfactual that the previous rung fails. Nonlinear dynamics is the stage on which computation happens; the primitives are the topology erected on it.

Three questions delimit the honest reach of the work. First, *necessity*: the paper establishes sufficiency chains inside one frozen family; it does not rule out other mechanisms that route, bind, or compose. Second, *generality*: every boundary wall is certified at frozen parameters, frozen queries, and two to four candidates; scaling audits (more distractors, nested operators, learned embeddings) are explicitly out of scope and listed in §10. Third, *composition’s open end*: the combiner tested is a single two-input algebraic node; what minimal organization suffices for multi-operand or nested composition is the natural next rung, and the paper leaves it unclaimed.

What the paper does claim, and defends with exact numbers, is a boundary map: for each rung from memory to composition, the minimal organizational condition under which the primitive appears in this substrate, and the exact statistic at which the previous rung stops. That map, not any single model, is the intended contribution.

10 Claim Ledger and Reproducibility Statement

This closing section grades every claim of the paper (Table 3) and states the reproducibility conditions under which every number in the preceding sections can be regenerated.

10.1 The claim ledger

Levels: ESTABLISHED (deterministic / analytical / numerical evidence, with exact statistics), SUPPORTED (statistical support without a closed-form theorem), REJECTED (explicitly negated by experiment), UNKNOWN (insufficient evidence in this programme).

#	Claim	Level	Frozen evidence	Anti-false-positive guard
C1	The scalar TAN has memory: its state is not Markov-sufficient	ESTABLISHED	$N = 160,000$; K mean: $0.5/4.47\text{e}3/1.235\text{e}4/2.047\text{e}4$; median: $0.5/896/1.37\text{e}3/2.95\text{e}3$; post-reset $0/0.6/33.1/32.2\%$	gate-closed collapse to 0.5; resync $n = 843$
C2	Positive-scalar saliency is order-locked (Theorem 5.1)	ESTABLISHED	FlipRate 0.000; Probe-2 TERMINATED/AUDIT_FAILED	domain-conditional statement (AM1)
C3	Attention reorganizes event-locked geometry without adding selectivity	ESTABLISHED	d_D 2.089 vs 1.484; $\Delta \log \text{Tr} = 1.355$ nats; $B4 \not\propto B1$	effective rank $\neq$ intrinsic dimension
C4	Nonmonotone tuning is opponent-structural; attention is gain / operating-range modulation	ESTABLISHED	shared-frame Theorem 4.1; C1 peak $x^* = 1.00$ ($h_E = 0.200$); C2–C4 no strict maxima; predictions 4/4	C1 has zero attention (necessity counterexample)
C5	Competition does not route	ESTABLISHED	six layouts $\times$ two variants FlipRate 0.000; query-sensitive magnitudes only	kernel-before-dynamics argument
C6	$d = 2$ is minimal for continuous query direction under dot-product; flip law θ/π	ESTABLISHED	132/132 isotropic conditions (140 checks total); $F_d = \frac{1}{2}$ across $d = 2, 3, 4, 8$; ordering ladder $1/2/M(M-1)/M!$	F_d retained as negative control; dimension controls
C7	The running neuron realizes counterfactual routing	ESTABLISHED	margins $+0.2587 \rightarrow -0.1559$; $Q^* = 3.7071$; ensemble 0.4425 vs 0.4445	probe-below-winner; ω -sweep structure; AM2 normalization
C8	Key–value decoupling yields soft, pairing-sensitive binding	ESTABLISHED	$T = -0.4129$; $T_{\text{swap}} = -T$ (10^{-12}); $D = 1.74/1.84$	value-swap antisymmetry; baselines (0.0844 sharpening drift)
C9	The WTA limit delivers exactly; routing $\perp$ discreteness	ESTABLISHED	γ ladder: $D \rightarrow 0$ only at ∞ ; flip rate flat 0.444527; E-I root $\gamma^* = 5.098015$	finite- γ positivity; AM5 cancellation
C10	Exact delivery does not imply composition (single-head bottleneck)	ESTABLISHED	$M0$ $E_{\text{comp}} = V_A > 0$; $V_A = 0$ trap caught by joint audit	zero flagged three-error joint audit
C11	Parallel binding plus one combiner composes; binding $\perp$ combiner error	ESTABLISHED	M2 errors 0 (S1–S3, $+/-$); both-routed cov. 22.03% (22.07% bench), $E_{\text{comp}} \text{both} \equiv 0$; bench 1.116/2.001, MC 1.108/1.976; 8 CFs pass	separation test; swap suite; M1 structurally undefined
C12	Biological readings (columns, phase, dendritic subunits) are organizational analogies	SUPPORTED (as mappings)	qualitative correspondence only	explicitly denied equivalence (rejected if read as identity)
C13	1D metric compatibility is sufficient for content addressing; 2D rotation not necessary	ESTABLISHED	60,000 trials (420,000 evals); C1/C2 acc 100.00%; metric $\equiv$ 2D argmax; signed scalar 0/6,705 on intermediate key	G1.1–G1.4 suites; chance derivation (16.67%); leakage audit
C14	Continuous dynamical carrier & spiking readout audit (Phase P1-B)	BOUNDED / INSUFFICIENT	96,000 episodes (192,000 queries); 6/6 cells at mutual floor (M1 joint delivery 2.75%–2.90% < 80%/40%); TOST 90% CIs bounded; spiking $E_{\text{comp}} \in [2.21, 2.57]$; abstention 82.1%–83.0%	pre-registered prerequisite floor rule; TOST equivalence bounds; G1.5 subthreshold collision theorem ($\text{res} < 10^{-10}$); H3 deferred; model-specific insufficiency

Table 3: Master claim ledger. Levels follow §2.5’s frozen discipline; every ESTABLISHED row carries its exact statistic and the audit that protects it.

10.2 Reproducibility statement

Archives. All quantitative statements of this paper are read from the frozen archives: the TAN-I dossier, collision and geometry logs, the Probe-1 amended audit, the Probe-2 identifiability archive, the TAN-II sprint registry 4.1–4.3-A (pre-registrations, amendments AM1–AM6, deterministic scripts, and machine-readable summaries), the Phase P1-A address identifiability archive (60,000 trials, 420,000 evaluations), and the Phase P1-B dynamical carrier audit dossier (Protocol V2.6 07D58A2D... , 12-fixture G1 suite, 96,000 episodes, confirmatory manifest 08EC79B9...). No frozen file was modified for this manuscript; the manuscript’s numbers were extracted, not recomputed.

Determinism. Every experiment is deterministic under the fixed seeds 20260904, 20260905, 20260906 ($N = 10^4$ per seed where Monte Carlo applies) with no random-state drift; analytic benchmarks are deterministic 2001² quadratures of the frozen model; structural identities are asserted at 10^{-12} ; the environment is Python 3.12 with NumPy/SciPy/Matplotlib only. Each sprint script records its own SHA-256 in its output summary.

Pre-registration and amendments. Every rung’s predictions were frozen before its run. Six discrepancies were found—AM1 (the E4(iii) closed-form value), AM2 (the unnormalized-control expectation), AM3 (corrected fixed-point constants), AM4 (the E-I drive-transfer value), AM5 (the readout-softness monotonicity and its cancellation zero), AM6 (the canonical routing margins)—and all six were handled by the frozen protocol: stop, independently recompute, record the cause in a dated amendment that preserves the original prediction verbatim, then continue. None altered a model, query, seed, or threshold; the originals remain in the archive.

Open problems (explicitly not claimed). General symbolic reasoning; multi-distractor ($N > 2$) generalization; necessity (versus sufficiency) of the tested mechanisms; nested or multi-operand composition; internal composition dynamics (Hypothesis H3, formally deferred); learned representations; and any biological equivalence (C12). These are the next rungs of the programme, and the paper’s boundary map is offered as the instrument for auditing them.

References

- [1] Shun-ichi Amari. Dynamics of pattern formation in lateral-inhibition type neural fields. *Biological Cybernetics*, 27(2):77–87, 1977.
- [2] Moses S. Charikar. Similarity estimation techniques from rounding algorithms. In *Proceedings of the Thirty-Fourth Annual ACM Symposium on Theory of Computing (STOC 2002)*, pages 380–388, 2002.
- [3] Jerry A. Fodor and Zenon W. Pylyshyn. Connectionism and cognitive architecture: A critical analysis. *Cognition*, 28(1–2):3–71, 1988.
- [4] Brenden M. Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, pages 2873–2882, 2018.
- [5] Brenden M. Lake, Tomer D. Ullman, Joshua B. Tenenbaum, and Samuel J. Gershman. Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40:e253, 2017.
- [6] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005.
- [7] André Martins and Ramón Astudillo. From softmax to sparsemax: A sparse model of attention and multi-label classification. In *Proceedings of the 33rd International Conference on Machine Learning (ICML 2016)*, pages 1614–1623, 2016.
- [8] Vernon B. Mountcastle. The columnar organization of the neocortex. *Brain*, 120(4):701–722, 1997.
- [9] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37(6):989–999, 2003.
- [10] Hubert Ramsauer, Bernhard Schäfl, Markus Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. *arXiv preprint arXiv:2008.02217*, 2020.
- [11] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*, pages 9355–9366, 2021.
- [12] Wolf Singer. Neuronal synchrony: A versatile code for the definition of relations? *Neuron*, 24(1):49–65, 1999.
- [13] Christoph von der Malsburg. Binding in models of perception and brain function. *Current Opinion in Neurobiology*, 5(4):520–526, 1995.